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Morality at the Box Office: A Narrative Economics Analysis, 1977–2025

Abstract

This thesis examines how the narrative structures of commercially successful films have changed over the past five decades. It uses synopsis and subtitle files as textual proxies for narrative content and analyzes the 20 highest-grossing films worldwide each year from 1977 to 2025 ($n = 980$) with a large language model. It introduces a framework, inspired by actor- and relation-based approaches in narrative economics, to classify and compare film narratives along a moral dimension. The framework captures two dimensions: the breadth of moral concern organized by film narratives and the geographic scale of the conflicts they depict. Across the sample window, both dimensions broaden substantially. The estimated trend corresponds to an increase of roughly 76% of the sample mean in the breadth of moral concern and about 120% in the geographic scale of conflict. These findings suggest that mainstream cinema's most commercially successful narratives have become broader in both moral and spatial terms.

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1. Introduction

This thesis studies how the narrative structure in commercially successful films changes over time. It focuses on the 20 highest-grossing films in each year from 1977 to 2025, treating them as the films with the greatest commercial reach in the theatrical market. Using plot synopsis and subtitle material as text-based proxies, the thesis analyzes film narratives with large language models (LLMs) to compare their structure across time. The thesis therefore studies the most economically successful titles in the theatrical market each year.

Existing research has identified many determinants of theatrical success, including reviews (Basuroy et al., 2003), word-of-mouth (Moretti, 2011), star power (Hofmann et al., 2017), sequels (Dhar et al., 2012), and release timing (Einav, 2007), but it has paid less attention to story content itself. Recent work suggests that moral and politically resonant content may matter for both commercial success and broader public impact (Gehring & Grigoletto, 2025; Huang et al., 2025; Michalopoulos & Rauh, 2024). Building on this literature, this thesis examines how commercially successful films organize harm, protection, and conflict over time. Rather than recovering the full political message of films, it studies their narrative structure through a politically relevant moral lens: the breadth of moral concern embedded in relations between hero, villain, and victim.

1.1 Characterization of the Theatrical Market

Because the empirical sample is drawn from the theatrical movie market, it is useful to first outline the basic economic characteristics of said market. The theatrical movie market is economically important, generating USD 42.3 billion in global revenue in 2019 (Motion Picture Association, 2022). However, the success of a particular film at the box office is notoriously difficult to predict. Films are experience goods (Nelson, 1970): audiences can only fully assess quality after consumption, while studios must commit production, marketing, and release decisions under substantial uncertainty about demand. Returns are heavy-tailed, with a small number of releases accounting for a disproportionate share of total revenues (De Vany & Walls, 1999). This unpredictability is reinforced by the seasonality of the theatrical market, in which a small number of release windows account for a disproportionate share of annual revenue (Einav, 2007).

The unpredictability of returns is also a symptom of experience-goods markets, in which the demand side partially compensates for uncertainty through the

transmission of information and signals. Such signals strongly shape audience beliefs through reviews and word-of-mouth, which can affect both early attendance and subsequent demand (Basuroy et al., 2003; Chintagunta et al., 2010; Reinstein & Snyder, 2005). Social influence and information cascades amplify small initial differences in performance into the heavily skewed revenue distributions characteristic of cultural markets (Cabral & Natividad, 2016; Delre et al., 2016). What gets amplified depends on what audiences and critics are willing to talk about and recommend, making story content an important success mechanism in theory. This mechanism works alongside supply-side signals such as star power, sequels, and release timing (Dhar et al., 2012; Einav, 2010; Elberse, 2007; Elberse & Eliashberg, 2003). Yet story content, let alone narrative structure, has received very little quantitative empirical attention.

1.2 Why Story Content May Matter

A small body of empirical work shows that story content can shape film outcomes. Huang et al. (2025) analyze approximately 1.6 million Internet Movie Database (IMDb) reviews and find that moral salience is strongly associated with film success. In particular, care, fairness, loyalty, and authority are positively related to ratings, popularity, and domestic box office revenue. Michalopoulos & Rauh (2024) document a related pattern at the level of plot. They show that the plots of commercially successful films often resemble long-standing local cultural narratives. Together, these findings indicate that story content provides an additional explanation for commercial and cultural traction, beyond the variables traditional film-industry research has identified.

A second line of work, in political economics, documents that film content is connected to political and social outcomes. Ang (2023) documents the effects of the film *The Birth of a Nation* (1915) on racial violence. Tan & Wang (2024), by contrast show that McCarthy-era repression shifted Hollywood film content in a more conservative direction. These studies point in different causal directions: films may shape social and political behavior, but political pressure may also shape film content. What they have in common is that measurable relationships appear where content is morally or politically resonant.

A natural first step is therefore to establish whether explicitly political content appears in commercially successful films and how its presence changes over time. A keyword-based descriptive benchmark on a broader top 100 sample shows that political topics are a recurring feature of commercially successful films across the

sample period, but their prevalence is not flat over time. Both the share of films containing at least one political keyword and the average within-film political keyword share rise from the late 1970s through the 1990s/2000s and decline thereafter. The top 20 sample tracks the top 100 closely while peaking earlier. Methodology, lexicon, and findings of this keyword analysis are reported in Appendix B.

1.3 From Topic to Narrative Structure

This first descriptive benchmark using keyword-based measures (reported in Appendix B) confirms that political topics are a recurring feature of commercially successful films across the sample period. Yet topical presence answers only the preliminary question of whether politics appears in a film. This thesis addresses a more substantive question: how films organize harm, protection, and conflict. To answer this question, the thesis moves from the topical presence of politics to narrative structure. Narrative is a loosely defined concept, allowing its operationalization to be tailored to the analysis at hand. However, this flexibility also makes the move from topics to narratives non-trivial and requires that a common understanding of the concept is established first.

The term narrative is used differently across disciplines and does not have a single settled definition. In narrative economics, narratives are not merely recurring themes, but socially shared stories that organize attention, interpretation, and ultimately behavior (Shiller, 2017). Recent syntheses similarly define economically relevant narratives as sense-making stories that emerge in a social context and implicitly suggest action, which helps distinguish them from topics or simple sentiment (Roos & Reccius, 2024). This understanding also connects to older work on persuasion and rhetoric in economics (McCloskey, 1998) and to political-science approaches that treat narratives as analyzable structures composed of characters, plots, and morals (Jones & McBeth, 2010). This thesis draws on both traditions: the narrative-economics view of narratives as socially shared, action-oriented stories, and the role-based political-science approach that treats narratives as structured configurations of characters and conflicts.

Beyond this conceptual challenge, a practical obstacle stands in the way: films are not natively textual. As audiovisual products, their narrative content is not directly machine-readable, which may help explain why little research bridges theatrical-market analysis and narrative economics. This obstacle is addressed through textual proxies, specifically plot synopsis and subtitle files. The advances in text-as-data

analysis have made narratives increasingly measurable through their structure: recent work codes actors, relations, and causal links from language and studies how variation in these narrative elements shapes observable outcomes (Ash et al., 2024; Bénabou et al., 2018; Eliaz & Spiegler, 2020; Gentzkow et al., 2019). Taken together, this literature suggests that narratives are best understood as socially shared and increasingly measurable structures that organize meaning, interpretation, and potential action.

This creates the central opening for the present thesis. The thesis asks who is threatened, who is protected, where conflict unfolds, and how these narrative structures change over time. In doing so, the thesis does not attempt to recover the full political message of a film, nor does it classify films as political or non-political in a binary sense. Instead, the findings are interpreted along a politically relevant moral dimension. The next chapter develops the conceptual framework used to analyze film narrative structure, introduces the moral dimension along which that structure is interpreted, and reviews the literature underpinning both.

2. Conceptual Framework

Large-scale quantitative analyses of narratives in film remain relatively rare, and existing work (Huang et al., 2025; Michalopoulos & Rauh, 2024) does not track narratives over time. By contrast, this thesis is concerned with how the narrative structure of commercially successful films changes over time. This focus requires a different analytical framework, where the growing economics text-as-data literature on measuring narratives in other domains (Ash et al., 2024; Gehring & Grigoletto, 2025) offers useful tools and concepts. The aim of this chapter is therefore to review the most relevant contributions from that literature and to develop the conceptual framework used in this thesis on that foundation.

2.1 Operationalization of Narratives

The broader economics literature offers more direct guidance on how narratives can be measured. A key starting point is Gentzkow et al. (2019), who show how language can be treated as quantitative evidence in economic research. Building on this foundation, recent work has operationalized narratives at different levels of structure. Ash et al. (2024) extract relational story structure from large text corpora, Eliaz & Spiegler (2020) model narratives as competing causal interpretations, and Bénabou et al. (2018) analyze narratives as moral justifications that guide and discipline behavior. While there is no gold standard, operationalizing narratives

through actors, relations, or evaluative links is well established and allows them to be coded systematically.

Role-based approaches are especially useful where narratives involve identifiable actors, explicit conflict, and clear evaluative roles. Political domains often display these features particularly clearly, which helps explain why political-science and political-economy research has been especially productive in developing such frameworks. In political science, the Narrative Policy Framework developed by Jones and McBeth (2010) treats narratives as analyzable structures composed of characters, plots and morals. Eliaz & Spiegler (2020) likewise highlight that politics is a domain in which competing stories can organize the same facts into different interpretations and policy conclusions. Most directly relevant for the present thesis, Gehring & Grigoletto (2025) define political narratives in role-based terms, combining a topic with identifiable characters and at least one assignment to the roles of hero, villain, or victim. This role-centered view also connects to broader narrative typologies such as Karpman's (1968) drama triangle and the wider hero–villain–victim tradition as discussed by Bergstrand & Jasper (2018). For the present application, role-based approaches provide a natural foundation for the analysis of film narratives over time.

2.2 Moral and Geographic Scope

A role-based account of narrative structure captures who appears in a story and how actors are positioned. It still leaves open an important question: along which dimension should these narratives be compared? Films involve many actors, settings, and topics at once, so a useful framework should reduce complexity without losing substantive political meaning. This thesis introduces two such dimensions: the moral scope (MS) of a narrative, which captures the breadth of moral concern it organizes, and the geographic scope (GS), which captures the spatial scale at which the central conflict unfolds.

The first dimension, moral scope, asks how broadly a narrative distributes harm, protection, and concern: whether concern remains narrowly oriented toward intimate or in-group others, or extends toward broader and more socially distant groups. This framing draws on the concept of moral universalism developed by Enke et al. (2023), defined as "the extent to which people exhibit the same level of altruism and trust towards strangers as towards in-group members" (p. 1924). Universalism is not a distinction between being more or less moral overall, but between concern that is concentrated narrowly toward in-group members and concern that is distributed more

evenly across social distance. Enke (2020) and Enke et al. (2023) show that moral universalism is systematically associated with broad clusters of policy preferences, ranging from redistribution, foreign aid, and environmental protection to border control, policing, and military spending. Moral scope therefore connects narrative structure to a politically meaningful dimension while reducing the broader moral landscape to a single interpretable question: how far concern extends.

The second dimension, geographic scope, captures the spatial scale at which the central conflict of a narrative unfolds. This dimension is necessary because the breadth of moral concern and the geographic scale of a story are related but not identical. A film may depict a highly global conflict while still organizing moral concern around a narrow in-group, or it may direct broad concern toward distant others while remaining locally bounded in spatial terms. Holding the two dimensions apart therefore matters. It allows the framework to distinguish narratives that are merely internationally scaled from narratives that are morally universalist in the sense of extending concern across social distance.

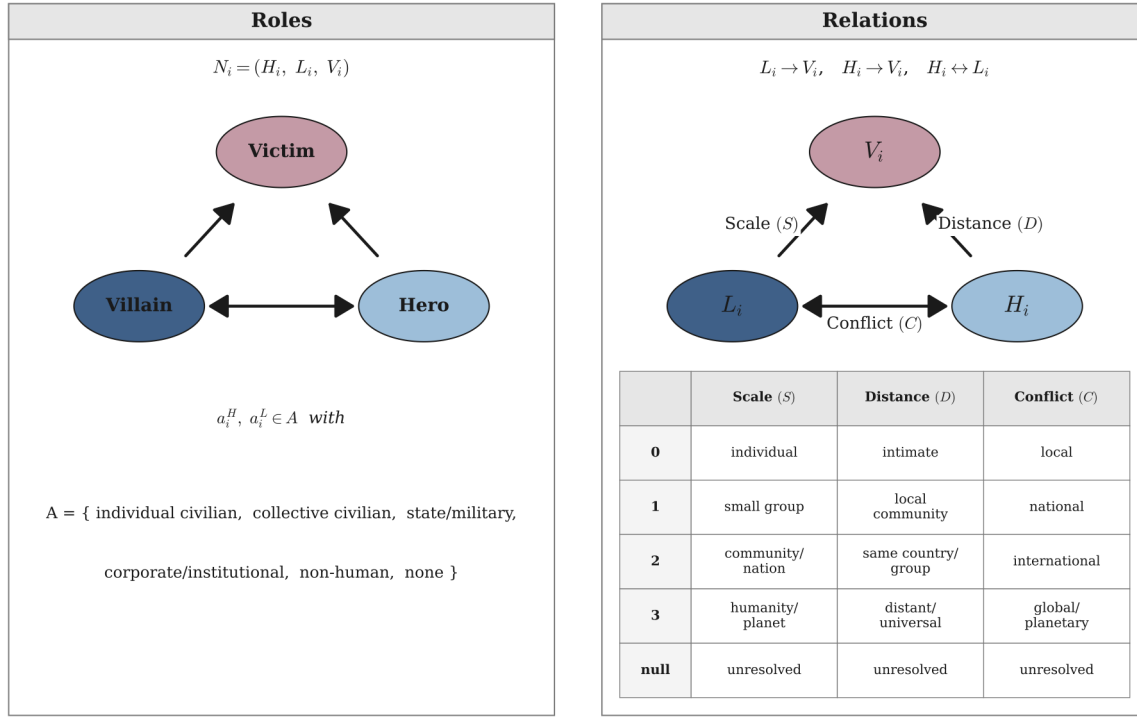
Operationally, this means analyzing the structure of narratives rather than interpreting their content. Following role-based work in narrative economics and political science introduced in Section 2.1, narratives are represented through roles and relations. Moral scope is then introduced as the interpretive lens through which variation in these structures is analyzed, not as the definition of the narrative object itself. The contribution of the thesis does not lie in identifying a film's complete ideological position, but in measuring two politically relevant dimensions of its narrative structure. Together, they allow the thesis to examine whether the narratives of commercially successful films become broader over time in moral terms, geographic terms, or both. The next section derives the relational coding framework used to operationalize both.

2.3 Relational Coding Framework

To operationalize the two dimensions introduced above, this thesis represents each film i as a role-structured narrative centered on three entities: a Hero (H_i), a Villain (L_i), and a Victim (V_i). Within this triad, only the Hero and the Villain are treated as actors in the strict sense, because they are the entities that act and thereby influence the outcome. If the victim itself became an active force driving the central conflict, it would also be coded as either Hero or Villain. Figure 1 summarizes the resulting framework consisting of the three roles, the actor-type set A and the three coded

ordinal relations Scale (S), Distance (D), and Conflict (C). The substantive level definitions of S, D, and C are introduced individually in the next paragraphs. The two derived measures, the Moral Scope Index (MSI) and Geographic Scope (GS), are then constructed from them.

Figure 1: Relational Coding Framework



Notes: The framework codes each film i as a Hero (H_i)–Villain (L_i)–Victim (V_i) triad $N_i = (H_i, L_i, V_i)$. Hero and villain receive actor-type assignments (a_i^H, a_i^L) from an Actor Set A . Three ordinal relations are defined: victim-group size S , hero–victim distance D , and conflict scale C , each ranging from 0 to 3. These yield the two main analysis measures: $MSI = S + D$ and $GS = C$. See main text for level definitions. Arrows mark actions and relations. Source: Own illustration.

First, the relation $L_i \rightarrow V_i$ captures the scale of the victim group that is hurt or threatened in the central conflict. The coding question is: What is the scale of the main threatened victim in the central conflict? This relation is coded as Scale (S). Second, the relation $H_i \rightarrow V_i$ captures how socially distant the main victim is from the hero. The coding question is: What is the social distance between hero and main victim, relative to the hero? This relation is coded as Distance (D). Third, the relation $H_i \leftrightarrow L_i$ captures the spatial scale at which the central conflict unfolds. The coding question is: What is the geographic scale of the central conflict? This relation is coded as Conflict (C).

Together, these three relational codings describe the structure of harm, protection, and conflict that organizes the film narrative. They are then used to derive the two main analytic dimensions of the thesis. The first is the Moral Scope Index (MSI), which

combines the scale of the harmed group with the social distance across which concern is extended. Formally,

$$MSI_i = S_i + D_i \quad (2.1)$$

where higher values indicate broader moral scope. The second dimension is geographic scope, which is captured directly by the coded conflict scale,

$$GS_i = C_i \quad (2.2)$$

If either S_i or D_i is unresolved, MSI_i is treated as missing. This construction translates the moral scope dimension introduced in Section 2.2 into a narrative setting.

Narratives are more universalist in this sense when they (i) depict protection or concern extending toward broader victim groups and (ii) across greater social distance between hero and victim. The framework therefore does not directly measure altruism or trust as defined by Enke et al. (2023). Rather, it captures a narrative analogue by observing how broad the victim group is, and how socially distant that group is from the hero. This offers a parsimonious way to compare film narratives across time. It reduces the complexity of films to a small set of comparable coded elements while preserving the central structure of harm, protection, and conflict.

To illustrate the full coding in one concrete example, consider *Harry Potter and the Deathly Hallows: Part 1* (2010). The heroes are Harry, Ron, and Hermione and the villains are Voldemort and the Death Eaters, so both actor-type codings fall into the collective civilian category. The victim is the wizarding world, which is broader than a single individual or a small group but narrower than all of humanity and thus a community/nation ($S = 2$). The hero collective belongs to the same wizarding community as the broader victim group, so the category same country or group is the best fit ($D = 2$). The central conflict unfolds primarily within Britain and within the institutions of the wizarding community, the conflict scale national applies ($C = 1$). The previously defined measures are therefore computed as shown in Equations (2.3) and (2.4).

$$MSI = S + D = 2 + 2 = 4 \quad (2.3)$$

$$GS = C = 1 \quad (2.4)$$

2.4 Expectations and Competing Arguments

The conceptual framework yields two central empirical questions. First, do commercially successful films increasingly organize harm and protection in broader moral terms over time? Second, do they increasingly organize conflict at broader geographic scales? These two dimensions are related but need not move together.

There are reasons to expect broader narratives over time. The theatrical market has become increasingly international, and the foreign share of global box office has risen as the U.S. share has declined (McKenzie, 2023). As commercially successful films become more dependent on foreign audiences, studios face stronger incentives to produce stories that travel across national boundaries. This argument suggests an expectation that commercially successful films may increasingly depict broader victim groups, greater social distance between hero and victim, and conflicts unfolding at larger geographic scales.

Broader moral concern is not guaranteed by market internationalization. Outward-facing orientations can take distinct moral forms: some root in broader concern for others, others in protection of the community and the defense of its boundaries (Kertzer et al., 2014; Rathbun et al., 2016). Enke et al. (2023) similarly distinguish universalist moral orientations from communitarian ones whose policy domains center on military spending and border control. A film can therefore depict global-scale conflict while still organizing protection around a narrow in-group. Commercial incentives may favor narratives that are geographically broader without necessarily becoming morally broader, which is precisely why the framework distinguishes the two.

The empirical analysis therefore does not assume that internationalization must produce broader narratives. It examines whether moral scope and geographic scope change over time, and whether such change is shared across both dimensions or concentrated in one. If moral scope broadens, the analysis decomposes the movement into its components, broader victim categories (S), greater social distance to those victims (D) or both. Geographic scope (C) is tracked separately to detect cases where the spatial reach of central conflicts changes independently of moral scope.

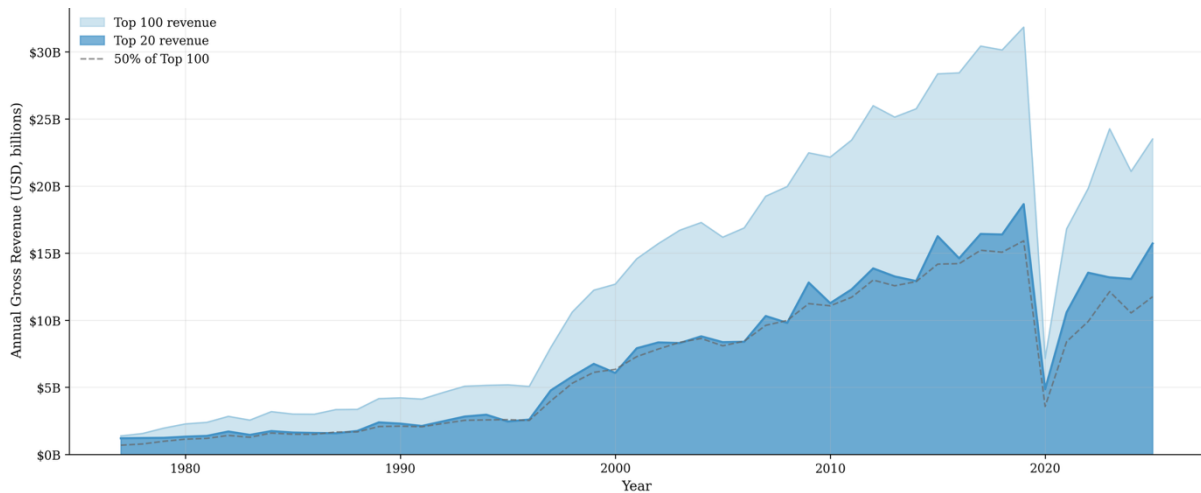
3. Data: Movie Sample 1977 – 2025

This chapter describes the empirical data used in the thesis. The main analytical dataset combines annual worldwide box-office rankings with movie identifiers, metadata, and text proxies for the top 20 highest-grossing films for every year from 1977 until 2025. The sample begins in 1977 which is the first year for which Box Office Mojo (n.d.) records worldwide gross figures. This sample provides the empirical basis for the narrative measures analyzed in the thesis. An auxiliary top 100 dataset is used for the keyword-based political-content benchmark in Appendix B.

3.1 Sample Construction and Market Coverage

The unit of observation is a film-year record. For each year from 1977 to 2025, the sample includes the top 20 films ranked by worldwide box-office gross. This choice targets the narratives with the highest potential cultural exposure each year and yields a balanced repeated cross-section with 20 observations per year ($N = 980$). To assess how concentrated annual revenues are among the highest-grossing titles, Figure 2 compares the total worldwide gross of the top 20 films to the total gross of the top 100 films in each year. The dotted line marks 50% of top 100 revenue. The figure shows that the top 20 films typically account for more than half of the top 100 revenue, capturing a large share of commercially dominant narratives each year.

Figure 2: Revenue Scale and Concentration in Top-Grossing Films, 1977–2025



Notes: Annual worldwide gross revenue in nominal USD billions. The darker area shows top 20 revenue, the full shaded area shows top 100 revenue. The dashed line marks 50% of top 100 revenue. Early top 100 coverage is partial for 1977–1979 and the 2020 drop reflects COVID-19 cinema closures. Source: Box Office Mojo (retrieved on 17.02.2026)

3.2 Data Sources

Box-office information from Box Office Mojo (n.d.) includes worldwide, domestic, and foreign gross (in USD) as well as derived shares and rank-based measures. Movies are matched to external identifiers from The Movie Database (TMDb, n.d.) and the Internet Movie Database (IMDb, n.d.) to retrieve metadata. The primary text proxy for narrative coding is an English plot synopsis, preferably from the Open Movie Database (Fritz, n.d.), with TMDb and Wikipedia (The Movie Database, n.d.; Wikimedia Foundation, n.d.) plot summaries as fallbacks when needed. Subtitles are sourced from OpenSubtitles (n.d.). The pipeline extracts context-rich evidence snippets from subtitles to support narrative coding decisions (documented in Appendix A.4).

3.3 Dataset Structure and Variables

Variables of the analysis dataset can be categorized into four groups: sample identifiers (e.g., film title, IMDb ID, release year), economic performance measures (e.g., domestic, foreign, and worldwide gross), text inputs (plot synopsis, subtitle file, TMDb keyword tags), and coded outputs (the LLM-coded narrative variables and the keyword-based political-content indicators). The full variable inventory with data types and sources is reported in Appendix D.1. Domestic refers to box-office revenue earned in the United States and Canada, while international (also called foreign) refers to revenue earned in all other theatrical markets. Worldwide is the sum of the two.

3.4 Data Quality and Coverage Diagnostics

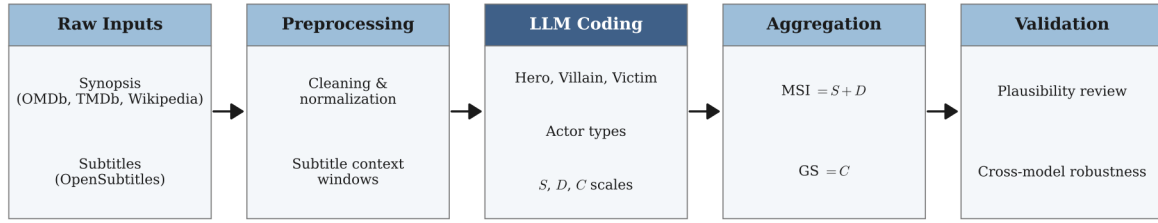
The main analytical sample is complete and balanced, with 20 films for each year from 1977 to 2025 ($N = 980$). Figures for sample structure, variable coverage, and missingness diagnostics are reported in Appendix D.1. Coverage of foreign box-office revenue is uneven across the sample window. Box Office Mojo, the source for annual rankings, notes that pre-2000 international totals reflect partial reporting, with some foreign revenues never consolidated into the worldwide totals (Box Office Mojo, 2005). Therefore, worldwide-gross figures for films from the late 1970s through the 1990s likely understate the foreign earnings of those films, particularly films with strong international audiences. Section 7.3 develops the analytical implications of this measurement issue.

4. Measurement and Pipeline

4.1 Overview of the Measurement Approach

Building on the dataset described in Chapter 3, this chapter explains how each film is converted into a coded narrative record. Figure 3 summarizes the pipeline: synopsis serve as the primary input for the coding model, while subtitles provide secondary context where the synopsis is short or ambiguous (subtitle preprocessing is documented in Appendix A.4). The pipeline combines deterministic text preprocessing with structured language-model coding and logs each step's parsing status for audit and replication. The remainder of the chapter proceeds in three sections: 4.2 specifies the coding schema, 4.3 documents how unresolved cases are treated, and 4.4 reports the validation.

Figure 3: Overview of the Measurement Pipeline.



Notes: The pipeline ingests two text inputs per film (a plot synopsis and a subtitle file) and converts them into a coded narrative record through three steps. Preprocessing applies deterministic cleaning, normalization, and subtitle-context-window extraction (see Figure A.1 for a more detailed visualization of the pipeline, Appendix A.4 for the full procedure). LLM Coding produces the role identifications, actor types, and the three ordinal scales S , D , C with a strict structured-output schema (Section 4.2 and Appendix A.5). Aggregation deterministically computes the composite measures $MSI = S + D$ and $GS = C$ from the model's outputs (Section 4.3 and Equations (2.1) – (2.2)). Validation comprises a manual plausibility review and a cross-model replication with an independent provider (Section 4.4 and Appendix C). Source: Own Illustration.

4.2 LLM Narrative Coding: Schema, Definitions, and Output Constraints

Narrative coding is performed with a strict structured-output schema at the film level. For each film, the model is asked to identify the hero, the villain, and the main threatened victim by name and to assign the actor type of the hero and the villain (six categories), the three ordinal scale variables (victim level, hero-victim distance, conflict scale), a self-reported confidence label, and a short structured reasoning summary. The full field-by-field schema with data types and allowed values is documented in Appendix A.5 and A.3. The three ordinal scale fields share a common 0–3 structure in which 0 denotes the narrowest scale and 3 the broadest. Their substantive level definitions are reported in Table 1.

Table 1: Definition of Ordinal Coding Levels for the Three Scale Variables

	0	1	2	3
S – Victim level	individual	small group	community or nation	Humanity or planet
D – Hero-victim distance	intimate	local community	same country or group	distant or universal
C – Conflict scale	local	national	international	global or planetary

Notes: S = victim_level (scale of the main threatened victim). D = hero_victim_distance (social distance between hero and victim). C = conflict_scale (geographic scale of the central conflict). All three variables additionally allow a null value when the available evidence is insufficient, see Section 4.3 for the null-handling policy. Source: author's coding.

Scale fields can also take the value null when the available evidence is insufficient to assign any of the four ordinal levels with confidence. This is a deliberate design choice rather than a measurement failure. The coding protocol permits unresolved outcomes and treats them as informative results, on the view that forcing a classification on weak evidence would inject noise into precisely the films where the narrative structure is genuinely ambiguous. The construction of the composite Moral Scope Index from the two underlying scales, and the broader treatment of unresolved cases at the analysis stage, are documented in Section 4.3.

4.3 Treatment of Unresolved Cases

The pipeline produces three ordinal variables at the film level: victim level (S), hero-victim distance (D), and conflict scale (C). The composite MSI, defined in Equation (2.1), is then constructed deterministically by the pipeline rather than produced by the language model itself. This way reliability depends only on the reliability of S and D. Two design questions then arise: when may the model return no value, and how is such a film handled in the resulting dataset?

The pipeline is allowed to return unresolved values when the available evidence is insufficient. Actor fields (`hero_actor`, `villain_actor`) can take the value `none`, and ordinal fields (S, D, C) can take `null`. These outcomes are treated as informative measurement results rather than coding errors. Imputation would inject artificial precision into a coding step whose entire premise is that some films do not have a clearly identifiable victim, hero, or conflict scale. The two unresolved categories carry different meanings. For actor fields, `none` is informative: a film with no clearly identified villain is structurally different from one with a named villain, and the frequency plots in Chapter 6 retain this category explicitly. For ordinal fields, `null` reflects insufficient evidence at the coding stage rather than a structural absence.

Across the 980-film sample, the rate of unresolved coding is low: 16 films are unresolved on S, 21 on D, 22 on C, and 21 on the MSI. In relative terms this means between 1.6 and 2.2 percent on any given variable. The MSI is constructed only when both S and D are observed for the same film. Figure D.1 in Appendix D.1 shows, for each of the six core narrative-coding outputs, the share of films with a usable value. How unresolved cases are then carried into the figures and regressions of Chapter 6 is an analysis decision documented in Section 4.3.

4.4 Validation

The narrative coding is validated in two steps. The first step is a manual plausibility review of a random subsample. The plausibility review checks whether the coded narratives are a coherent reading of the source material the model was given. The second step is a full replication of the coding with an independent model from a different provider. This shows to what degree the findings depend on which model is used as the coder.

4.4.1 Plausibility Review

A 10% subsample of the 980 films ($n = 98$) was drawn for manual review by randomly selecting two films from each year, using a fixed seed (42) for reproducibility. Each case was reviewed in a terminal interface that showed the film's metadata, its synopsis, the full model output (hero, villain, victim role, ordinal codes, per-field reasoning, and the model's own confidence), and the parsing status. The reviewer marked each case as plausible or implausible based on whether the coded narrative made sense given the synopsis and, where present, the subtitle evidence. Decisions were saved after every case, and a session report was produced (see Appendix C and the run log of the Python project).

Of the 98 cases, 84 (85.7%) were judged plausible and 14 (14.3%) implausible. Implausibility is not spread evenly across confidence levels output by the model. None of the 32 high-confidence cases was judged implausible, while 10 of 57 medium-confidence cases (17.5%) and 4 of 9 low-confidence cases (44.4%) were. This clear gradient matters in two ways. First, it shows that the model's own confidence flag is a real signal. Second, this supports the principle established in Section 4.3 of allowing unresolved outputs rather than imputing values. The cases where the model self-flags low confidence are the same cases most likely to be coded wrong, and forcing a definitive answer would increase errors.

4.4.2 Cross-Model Robustness

The primary coder in the analysis is GPT-5.4 (OpenAI). To test whether the trends depend on the model used in the primary run, the full coding was repeated on the same sample ($n = 980$) with Claude Opus 4.6, a model from a different provider (Anthropic). Both runs used the same prompts, preprocessing, output schema and parsing rules (Appendix A). The only intended difference was the coding model itself.

Agreement between the two runs is reported at the item level in Table 2. Item-level agreement is substantial across all variables (Cohen's $\kappa \geq 0.60$ throughout, in the

substantial-agreement range of Landis & Koch (1977)). Hero and villain actor categories match exactly in 82% and 68% of cases respectively, and the three ordinal components reach exact-match rates of 74–80% with within-one-rank agreement of 91–94%. The composite MSI shows the lowest exact-match rate (63%) due to the accumulation of component disagreements.

Table 2: Cross-Model Item-Level Agreement (GPT-5.4 and Claude Opus 4.6)

	(1) n	(2) Exact match	(3) Within-one	(4) κ
Panel A: Nominal role variables (Cohen's κ)				
hero_actor	979	82.4%	—	0.707
villain_actor	979	67.6%	—	0.602
Panel B: Ordinal scale variables (quadratic-weighted κ)				
S — Victim level	940	74.4%	93.6%	0.774
D — Hero-victim distance	936	74.3%	90.7%	0.745
C — Conflict scale	954	80.3%	92.2%	0.791
MSI — Moral Scope Index	936	62.8%	82.9%	0.798

Notes: Each row reports item-level agreement between the primary coding run (GPT-5.4) and the secondary coding run (Claude Opus 4.6) on the matched sample. n is the number of films with a valid (non-null) code in both runs. Exact match is the share of films coded identically. Within-one is the share of films whose codes differ by at most one rank and is reported only for ordinal variables. Cohen's κ (Panel A) and quadratic-weighted κ (Panel B) adjust the agreement rate for chance given the marginal distributions. Values of 0.61–0.80 indicate substantial agreement (Landis & Koch, 1977). MSI agreement is mechanically lower than its components because errors in S and D compound. Source: Own calculations based on GPT-5.4 and Claude Opus 4.6 coding runs.

At the aggregate level, Figures C.1-C.4 show visually that the year-by-year means of the two runs move in the same direction and peak at similar times across the scale variables S, D, C, and the MSI. Claude Opus 4.6 codes slightly higher on average than GPT-5.4 across all four variables (Table C.2). The absolute level of the coding is therefore somewhat model-dependent, but the trend over time is not. Together with the plausibility review in Section 4.4.1, this cross-model agreement supports interpreting the trends in Chapter 6 as features of the underlying data rather than artefacts of a specific coding model.

5. Analytical Approach

5.1 Analytical Design

This chapter outlines how the thesis evaluates the hypotheses using descriptive evidence from the film-level narrative coding. The final dataset is a balanced repeated cross-section of the annual top 20 worldwide box-office films from 1977 to 2025, with the film-year record as the unit of observation. The empirical objective is not to identify a causal effect, but to examine whether the narrative scope of commercially successful films changes systematically over time.

Given this design, the analysis relies primarily on descriptive comparisons across time. This thesis uses film-level coded outcomes to study temporal developments in narrative structure and to compare the composition of coded categories across the sample. The main evidence is presented graphically, using yearly or binned averages for ordinal scale variables and frequency-based visualizations for categorical role configurations. These graphical summaries are complemented by simple ordinary least squares (OLS) trend regressions of each ordinal outcome on the calendar year, which quantify the magnitude and statistical precision of any observed change. The regression specification, the choice of standard errors, and the rationale for not including control variables are detailed in Section 5.3.

Unresolved cases (Section 4.3) enter the analysis differently depending on the output. In frequency plots, unresolved categories are shown explicitly where relevant. A film coded `villain_actor = none` is reported as such rather than dropped, since the absence of a clear antagonist is itself substantive. In yearly means and trend regressions, films with a missing value on the outcome variable are excluded without imputation. Given that fewer than 2.5% of the sample is unresolved on any given variable, the bias from deletion is small relative to the magnitudes of the trends documented in Chapter 6.

5.2 Mapping Research Questions to Outcome Variables

The research questions developed in Sections 2.2 and 2.4 map directly onto the outcome variables from the coding framework. Whether moral scope has broadened over time is evaluated using the MSI, with its two components examined separately: S captures the scale of the main threatened victim, while D captures the social distance between the hero and that victim. The geographic dimension is evaluated using C, which captures the scale of the central conflict and is tracked separately from moral scope.

5.3 Trend Estimation and Scope of Interpretation

This thesis estimates time trends using a linear regression of each coded narrative outcome on the calendar year. The primary specification is estimated at the film level:

$$y_{i,t} = \beta_1 + \beta_2 \cdot \text{year}_t + \varepsilon_{i,t} \quad (5.1)$$

where $y_{i,t}$ denotes the coded narrative outcome for film i in year t , with $y \in \{MSI, S, D, C\}$. The coefficient β_2 captures the average annual change in outcome over the sample period $t = 1977, \dots, 2025$. The error term $\varepsilon_{i,t}$ absorbs all film-specific and time-specific factors that are not modelled explicitly. Films with a missing value for the outcome variable are excluded from the respective regression. Standard errors are clustered by year to account for common market and industry conditions affecting the 20 films in each annual cross-section. With 49 year clusters, cluster-robust inference is appropriate for this descriptive specification (Colin Cameron & Miller, 2015).

The specification does not include control variables, because the research question concerns how commercially successful film narratives evolve over time, and shifts in genres, franchises, or production modes are part of the phenomenon under study rather than confounders to be partialled out. A specification that controlled for genre would estimate how narratives shift within the same genre category, which is a different empirical question from the one posed in Chapter 1 and 2. Practically, commercial films typically belong to multiple, overlapping genres, and genre taxonomies themselves have shifted over the 49-year window, which makes a clean within-genre fixed-effects specification difficult to justify in this data context.

The coded outcomes are ordinal at the film level, with S , D , and C taking integer values 0–3, and MSI ranging over 0–6 as the sum of two ordinal scales. OLS is used as a linear approximation for trend estimation, and the coefficient β_2 is interpreted as the average per-year change in the outcome. As a robustness check, the same regression is also estimated on the 49 yearly means:

$$\bar{y}_t = \beta_1 + \beta_2 \cdot \text{year}_t + \varepsilon_t \quad (5.2)$$

Where $\bar{y}_t$ denotes the mean value of the respective coded outcome across all films observed in year t .

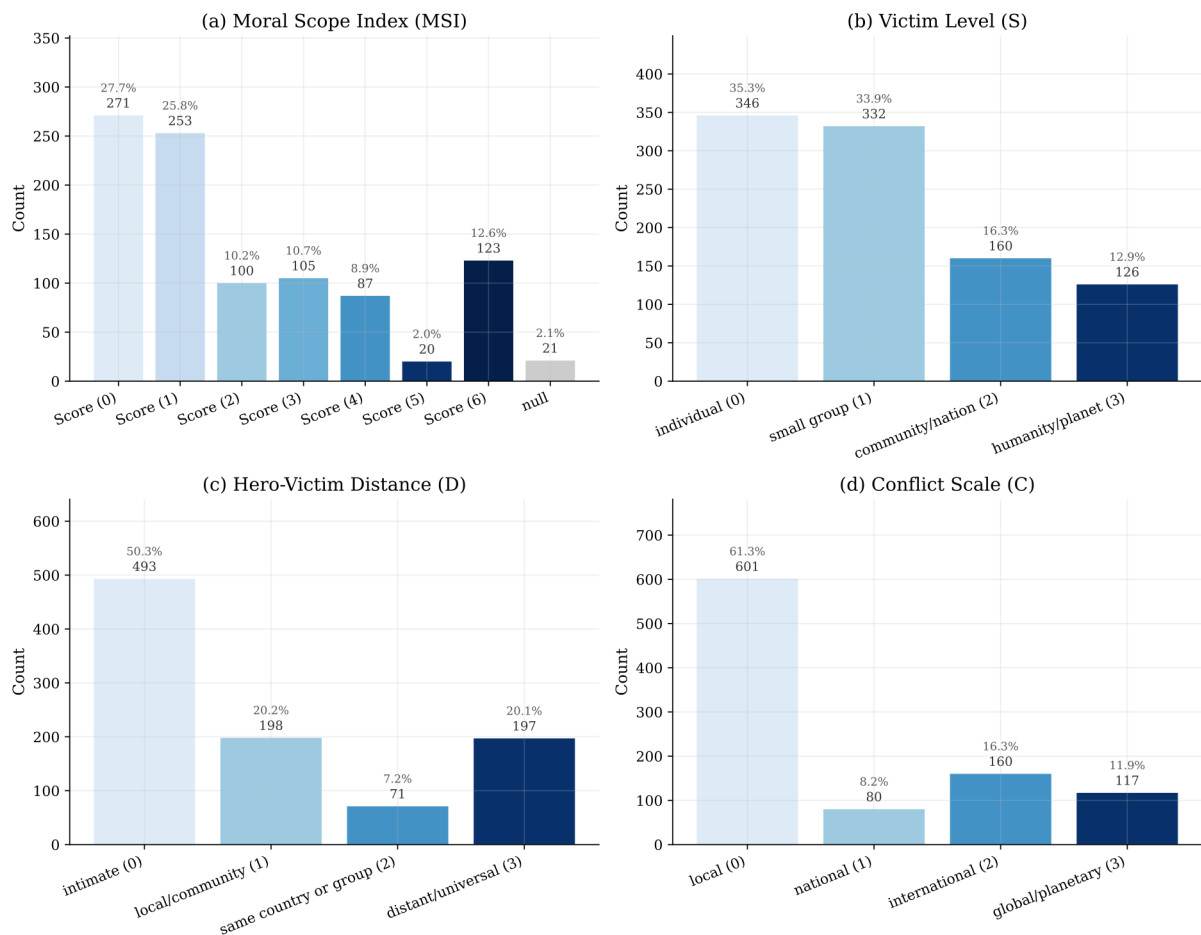
Both specifications are reported in Table 3 in Section 6.5. Coefficient estimates from the two specifications are effectively identical, showing that the trend is robust to the level of aggregation. The analysis is descriptive rather than causal. The estimated time trend documents whether commercially successful films have become broader in moral or geographic scope over the sample period. It does not, by itself, establish why such changes occur.

6. Results

6.1 Distributional Overview

This section describes the pooled distributions of the Moral Scope Index (MSI), its two components (S and D), and the geographic scope variable (C), as well as the hero- and villain-actor categories before turning to trends. Figure 4 displays the distributions of all four narrative-scope variables. Visually the MSI is right-skewed. While S is more uniform across its lower half, D is far more concentrated at zero, leading to the right-skew in the MSI. C is even more concentrated at zero, nevertheless S, D and C all carry a non-trivial top tail.

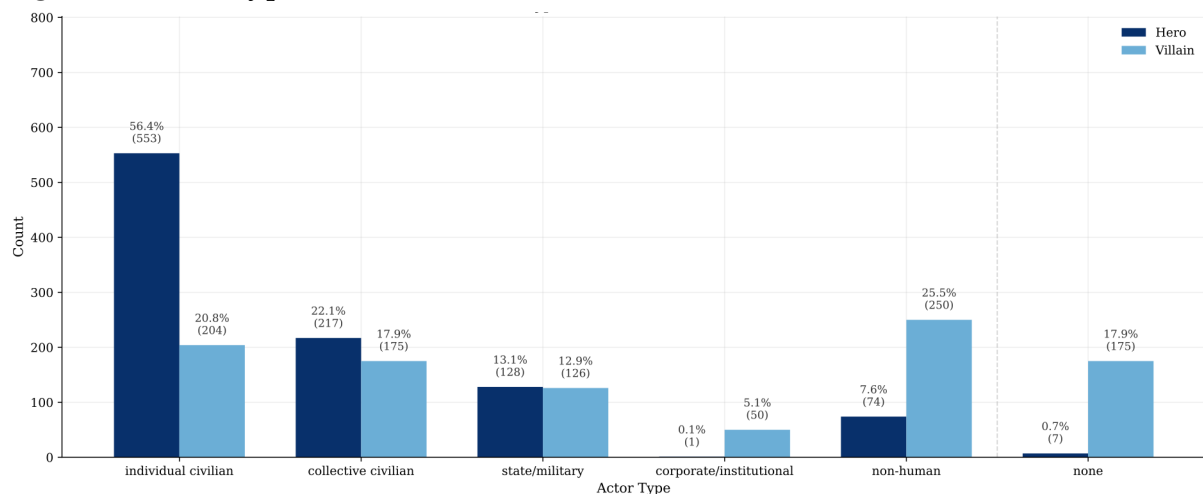
Figure 4: Pooled Distributions of the Moral Scope Index and the Three Coded Relations, 1977–2025.



Notes: Sample: the 20 highest-grossing films worldwide per calendar year, 1977–2025 ($n = 980$). In Panel (a), unresolved cases ($n = 21$, 2.1%), where either S or D is missing, are shown in the "null" category (see Section 4.3 for the null-handling policy). Panels (b)–(d) omit unresolved cases ($n = 16$ for S, 21 for D, 22 for C). Panel (a) shows the Moral Scope Index ($MSI = S + D$, range 0–6). Panel (b) shows Victim Level (S, range 0–3). Panel (c) shows Hero-Victim Distance (D, range 0–3). Panel (d) shows Conflict Scale (C, range 0–3), the measure of geographic scope. Source: Own calculations based on the GPT-5.4 narrative-coding outputs for the top 20 worldwide box-office sample, 1977–2025.

Figure 5 shows the actor-type distributions of hero and villain roles separately. The hero role is dominated by individual civilians: 56% of films cast a single civilian protagonist, while corporate or institutional heroes are essentially absent, appearing in only one film. Villains are distributed more evenly across actor types. Non-human antagonists form the largest single category at 26%, followed by individual civilians, collective civilians, villains coded as none, state or military villains, and corporate or institutional villains. The asymmetry is especially visible in the none category, which appears 174 times on the villain side but only six times on the hero side. This suggests that films often have a clearly identifiable protagonist, but not necessarily a clearly identifiable antagonist.

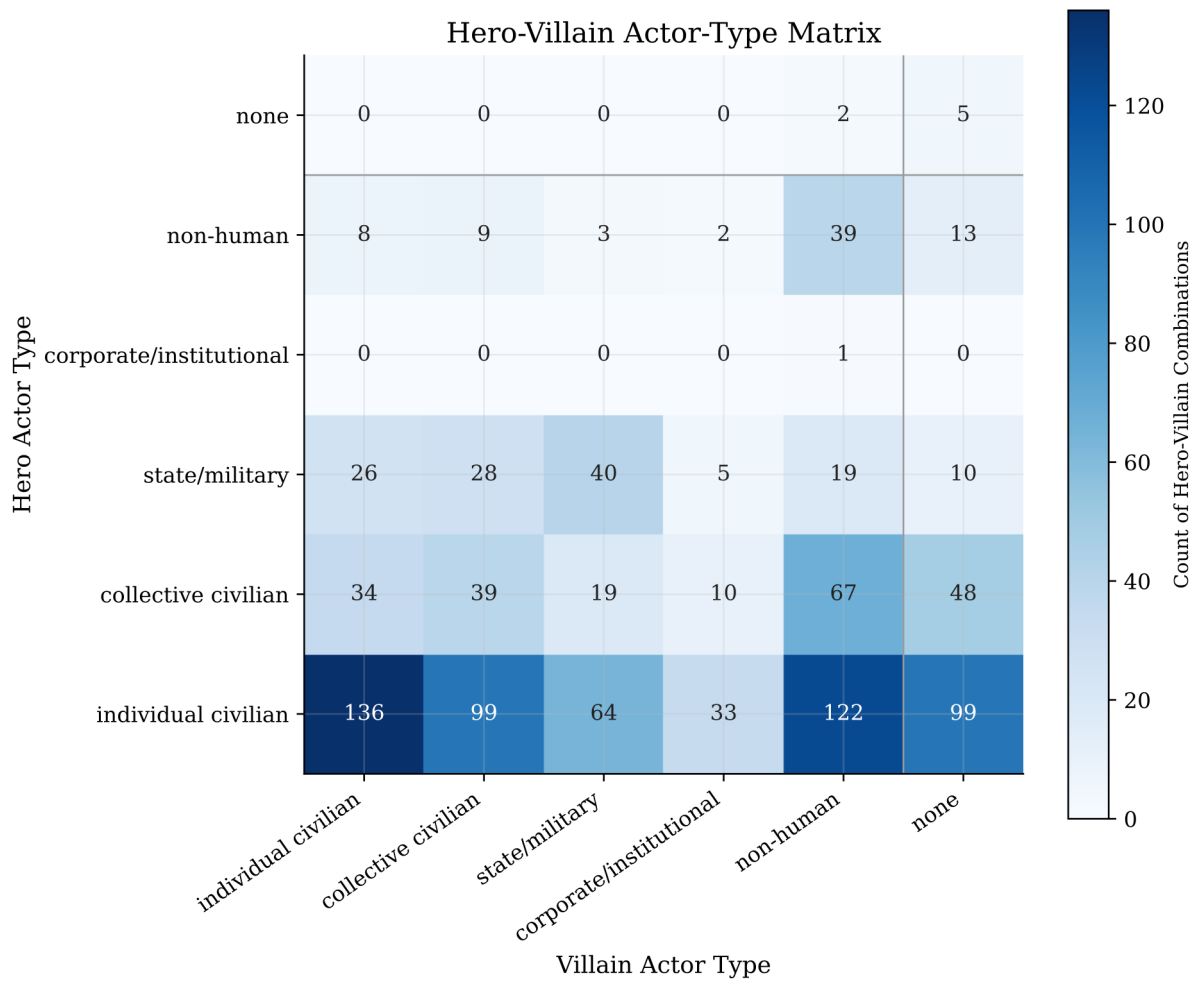
Figure 5: Actor-Type Distribution for Hero and Villain roles, 1977–2025



Notes: Pooled actor-type distribution for hero and villain roles (n = 980 films). The five substantive actor-type categories appear to the left of the dashed line. The none category to the right marks films in which no actor or role could be identified. See Section 2.3 for role-identification and Section 4.3 for null handling. Source: Own calculations based on the GPT-5.4 narrative-coding outputs for the top 20 worldwide box-office sample, 1977–2025.

Figure 6 shows the joint distribution of hero–villain actor-type pairings. The heatmap confirms the asymmetry between hero and villain roles at the level of pairings. The most frequent combinations are concentrated around individual-civilian heroes, reflecting the strong dominance of this category in the hero role. The four most common pairings all involve an individual-civilian hero and together account for roughly half of the sample. By contrast, villain roles are spread more widely across categories. The joint distribution therefore shows that top-grossing films tend to center individual heroes while allowing for a much broader range of villains in their narrative structures.

Figure 6: Joint Distribution of Hero and Villain Actor Types, 1977–2025

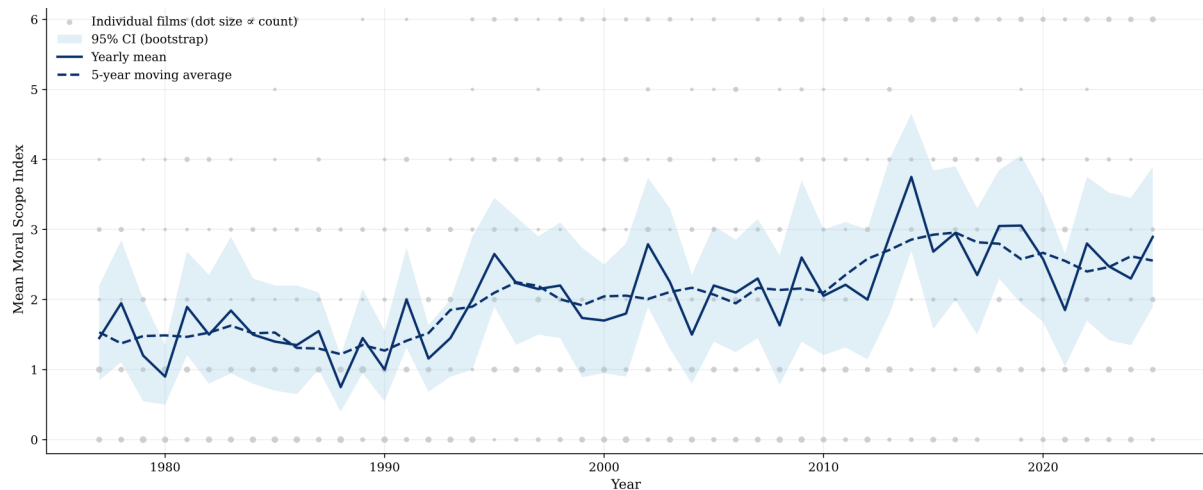


Notes: Cell counts report the number of films with the corresponding hero × villain actor-type combination across the full sample (20 highest-grossing films worldwide per calendar year, 1977–2025, $n = 980$). Row and column marginals reproduce the hero and villain distributions shown in Figure 5. The none row and column retain films in which the corresponding role could not be identified, see Section 2.3 for the role-identification convention. Darker shading indicates higher frequency. Source: Own calculations based on the GPT-5.4 narrative-coding outputs for the top 20 worldwide box-office sample, 1977–2025.

6.2 Moral Scope Over Time

The pooled picture in Section 6.1 ignores time. This section therefore asks whether the moral scope of top-grossing films has shifted over the 49-year observation window. The central outcome is the Moral Scope Index, which combines the scale of the threatened victim group with the social distance between hero and victim. A rise in the MSI indicates that successful film narratives increasingly organize harm and protection around broader or more socially distant victims. Figure 7 plots the yearly mean of the MSI from 1977 to 2025, together with a 5-year moving average and 95% bootstrap confidence intervals. Table 3, reported at the end of this chapter, then summarizes the corresponding regressions for all four scope variables.

Figure 7: Yearly Mean Moral Scope Index, 1977–2025



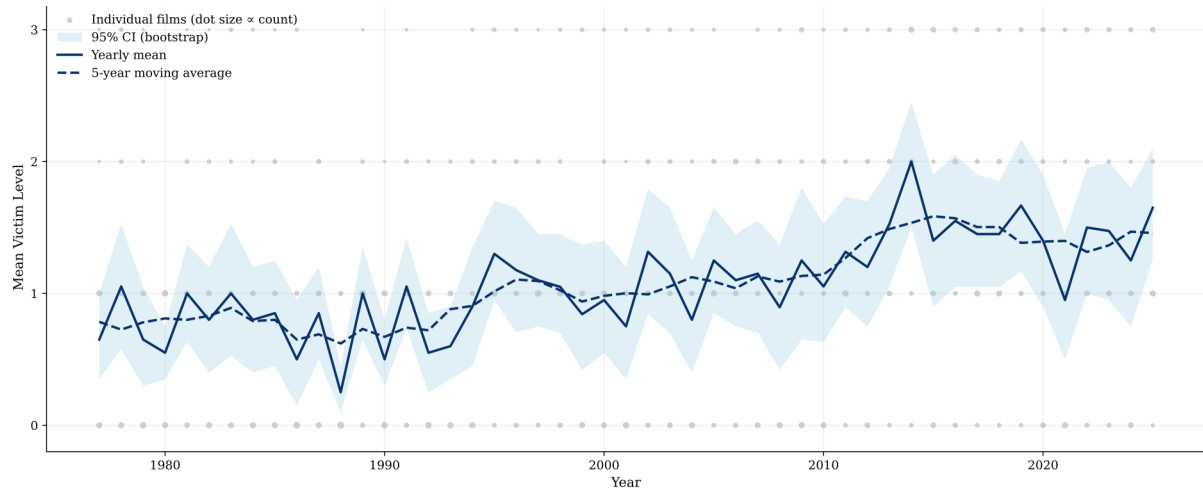
Notes: Yearly cross-sectional mean of the MSI (range 0–6) across the 20 highest-grossing films per year, 1977–2025 ($n = 959$, films with unresolved S or D excluded). Solid line: yearly mean. Dashed line: 5-year centered moving average. Shaded band: 95% bootstrap CI. Grey markers show distinct MSI values present each year, sized by film count. The sample mean of the MSI is 2.04 ($n = 959$). Source: author's coding and calculations.

Figure 7 shows a clear upward drift. The yearly MSI mean roughly doubles across the window, from about 1.5 in the late 1970s to about 3 in 2025. The 1980s and early 1990s contain several years below the sample mean of 2.04. The 2010s and 2020s sit mostly above it. Year-to-year variation is substantial, but no single year dominates the picture, and the slope appears linear rather than shaped by a discrete break. Over the full window the trend amounts to a rise of roughly 76% of the sample mean, a small per-year movement that accumulates into a substantial overall shift in how the most successful films distribute moral concern.

6.3 Decomposing Moral Scope

The rising MSI in Section 6.2 could be driven by the component S, the component D, or by both. This section separates the two. Figure 8 plots yearly means of S and Figure 9 plots yearly means of D, with a 5-year moving average and 95% bootstrap confidence intervals. Figure 8 shows a steady rise in S. Decade means roughly double from the 1970s to the 2010s before levelling off, with moderate year-to-year variation and no discrete break. Over the 49-year window the estimated rise is about 83% of the sample mean. Substantively, top-grossing films have shifted away from stories in which only a named individual is at stake, toward stories in which larger collectives, or humanity as a whole, are at stake.

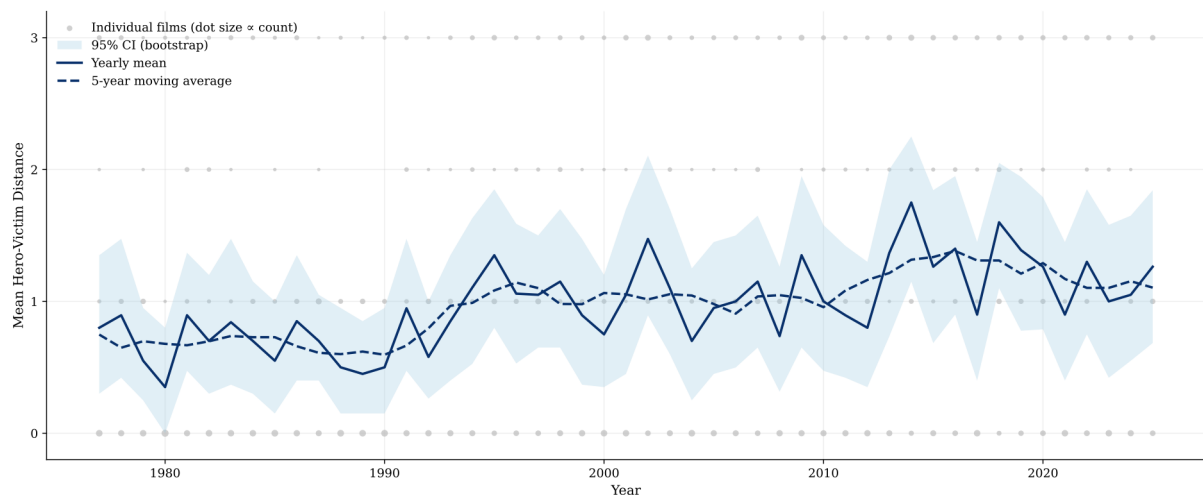
Figure 8: Yearly Mean Victim Level (S), 1977–2025



Notes: Yearly cross-sectional mean of victim level S (0 = individual, 1 = small group, 2 = community/nation, 3 = humanity/planet) across the 20 highest-grossing films per year, 1977–2025 (n = 964, unresolved S excluded). Solid line: yearly mean. Dashed line: 5-year centered moving average. Shaded band: 95% bootstrap CI. Grey markers show distinct S values present each year, sized by film count. The sample mean of S is 1.07 (n = 964). Source: author's coding and calculations.

Figure 9 shows a similar upward path for D, though the slope is flatter and the series noisier. Decade means rise from the 1970s into the 2010s, then dip slightly in the 2020s. Over the full window the rise amounts to about 69% of the mean. Substantively, contemporary top-grossing films more often place the hero at social or institutional distance from the victim, rather than the victim being a part of close family, friends, or immediate community.

Figure 9: Yearly Mean Hero-Victim Distance (D), 1977–2025



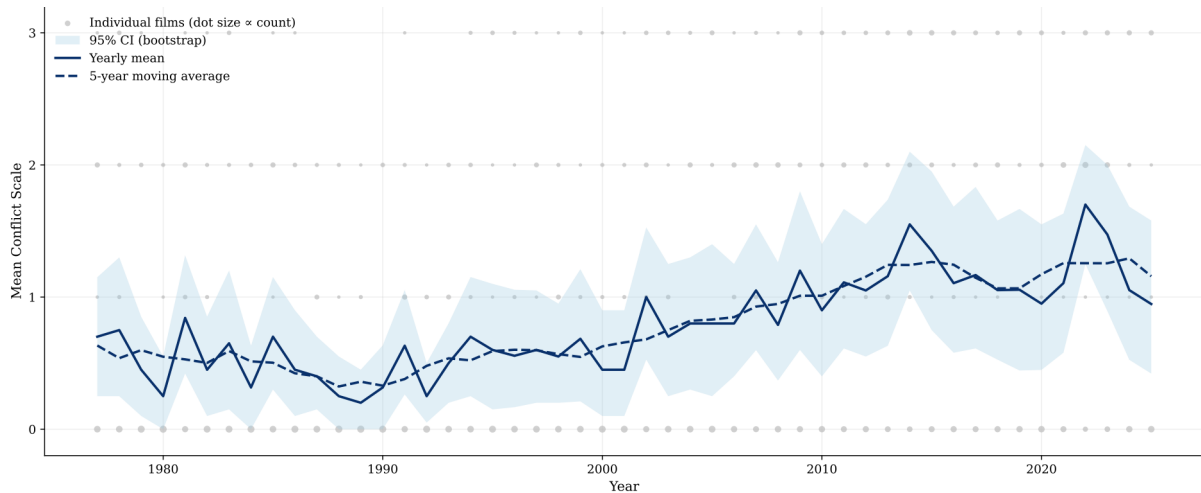
Notes: Yearly cross-sectional mean of hero-victim distance D (0 = intimate, 1 = local community, 2 = same country/group, 3 = distant/universal) across the 20 highest-grossing films per year, 1977–2025 (n = 959, unresolved D excluded). Solid line: yearly mean. Dashed line: 5-year centered moving average. Shaded band: 95% bootstrap CI. Grey markers show distinct D values present each year, sized by film count. The sample mean of D is 0.97 (n = 959). Source: author's coding and calculations.

The two components together reproduce the MSI trend by construction, and the decomposition shows that neither dominates. S contributes about 57% of the 1.58-point rise in the MSI over the period, D about 43%. Both are precisely estimated (Section 6.5) and large relative to their means, but S accounts for somewhat more of the movement. Section 6.4 turns to geographic scope, which has been tracked separately from the MSI.

6.4 Geographic Scope Over Time

The variable C captures the geographic reach of the story's core conflict, independently of who is at stake. It is tracked separately from the MSI to keep the moral-scope composite focused on stakes (S) and hero-victim distance (D). Figure 10 plots the yearly mean of C from 1977 to 2025, with a 5-year moving average and 95% bootstrap confidence intervals.

Figure 10: Yearly Mean Conflict Scale (C), 1977–2025



Notes: Yearly cross-sectional mean of conflict scale C (0 = local, 1 = national, 2 = international, 3 = global/planetary) across the 20 highest-grossing films per year, 1977–2025 (n = 958, unresolved C excluded). Solid line: yearly mean. Dashed line: 5-year centered moving average. Shaded band: 95% bootstrap CI. Grey markers show distinct C values present each year, sized by film count. The sample mean of C is 0.78 (n = 958). Source: author's coding and calculations.

Figure 10 shows a clear but non-monotonic rise, with most of the increase occurring after 1990. The yearly series remains noisy, but the corresponding trend regression in Table 3 indicates a positive average change across the full window. Over the 49-year period the estimated rise amounts to about 120% of the sample mean, making it the largest relative movement of the four scope variables. The composition of C over time, available in Appendix D.2, shows that the increase is driven primarily by a shift away from local conflicts (C = 0) toward international and global scales (C = 2, C = 3) rather than national-scale stories (C = 1).

6.5 Trend Regression

The graphical evidence in Sections 6.2–6.4 is quantified by simple OLS regressions of each ordinal outcome on the calendar year (details in Chapter 5). Table 3 reports the resulting trend estimates for all four narrative-scope variables, with standard errors clustered at the year level (Panel A) and on the 49 yearly means as a robustness check (Panel B). All four trends are positive and statistically significant at conventional levels. The film-level slope for the MSI is 0.0323 per year (SE 0.0044, $p < 0.001$), implying a rise of about 1.55 points across the 49-year window, roughly 76% of the sample mean. For the components, S rises at 0.0186 per year (overall rise of ≈ 0.89 points, 83% of the mean), D at 0.0139 (overall rise of ≈ 0.67 points, 69% of the mean). Geographic scope C shows the largest relative movement: a slope of 0.0195 implies a rise of 0.94 points over the window, 120% of its sample mean.

Table 3: OLS Trend Estimates for Narrative Structure Variables, 1977–2025

	(1)	(2)	(3)	(4)
	MSI	S	D	C
Panel A: Film-level OLS, cluster-robust SE				
on year				
Year	0.0323 (0.0044) [<0.001]	0.0186 (0.0024) [<0.001]	0.0139 (0.0023) [<0.001]	0.0195 (0.0027) [<0.001]
R ²	0.051	0.067	0.027	0.061
Observations (N)	959	964	959	958
Mean of dep. var.	2.04	1.07	0.97	0.78
Panel B: Yearly-means OLS (robustness, N = 49)				
Year	0.0322 (0.0045) [<0.001]	0.0185 (0.0025) [<0.001]	0.0138 (0.0025) [<0.001]	0.0194 (0.0023) [<0.001]
R ²	0.519	0.546	0.398	0.602
Observations (years)	49	49	49	49

Notes: Standard errors in parentheses, p-values in square brackets. Panel A uses film-level OLS with standard errors clustered on year. Panel B regresses yearly means on year with heteroskedasticity-robust standard errors. Dependent variables: MSI = Moral Scope Index (0–6), S = Victim Level (0–3), D = Hero-Victim Distance (0–3), C = Geographic Scope (0–3). Sample: top 20 films by worldwide box office, 1977–2025 (Observations (N) shown in Table). Source: Own estimation on master_films_stage4_narrative_top20_gpt54.

Panel B confirms that the trends are not driven by within-year heterogeneity: the yearly-means specification returns nearly identical slopes. The R^2 values are small (0.027–0.067 in Panel A), reflecting that a linear year trend captures changes in the central tendency over time and does not explain individual film-level variation. The fits are nevertheless precise enough to distinguish all four slopes from zero at conventional significance levels. Section 5.3 cautions that linear fits on ordinal composites should be read as descriptive trends rather than causal estimates.

Taken together, Chapter 6 produces four consistent findings. First, top-grossing films are concentrated at the lower end of all four scales in the pooled sample. Second, the MSI has risen by roughly 1.55 points across the 49-year window, about 76% of its sample mean. Third, both components of the MSI share in that movement, with S contributing somewhat more than D. Fourth, geographic scope has moved in the same direction as S and D, and by even more in proportional terms. Chapter 7 turns from documenting these patterns to interpreting them: what might explain the shift, what its limits are, and how it connects to the broader question of how mainstream cinema represents moral and geographic scope.

7. Discussion and Limitations

The trends documented in Chapter 6 unfold within the unpredictable theatrical market described in Section 1.1. That context should be considered when interpreting the findings. Films are experience goods whose quality cannot be assessed before consumption, returns are heavy-tailed, and small early differences in performance can be amplified into large outcome gaps (De Vany & Walls, 1999; Einav, 2007). The films that enter the annual top 20 are not chosen by studios or consumers in isolation. It is a product that needs to spread rapidly, sustained by word-of-mouth, and thereby achieve a kind of virality that wins the attention of mass audiences. The narratives studied in this thesis are therefore the winners of that competition for attention, and that determines what the trend can be read to mean and what it cannot.

7.1 Interpreting the Trend

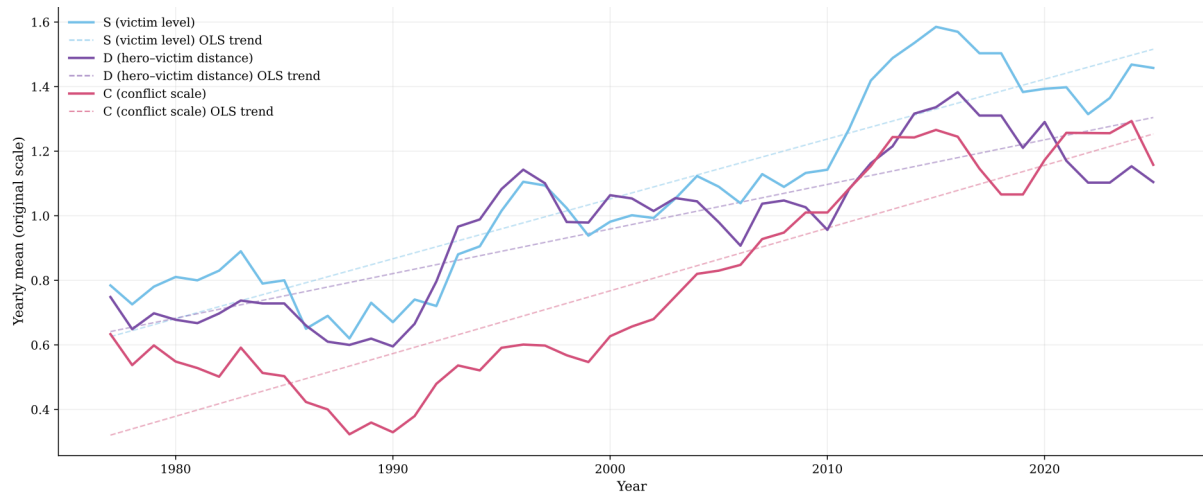
Films that enter the annual top 20 win in a competition for attention shaped by uncertainty, word-of-mouth, and social influence (Basuroy et al., 2003; Cabral & Natividad, 2016; Delre et al., 2016). In the narrative-economics view of Section 1.3, they can be read as stories that achieved unusually broad commercial traction. The trends documented here show which kinds of moral and geographic narratives appear

among these commercially viral films over the past five decades. They cannot identify whether studios, audiences, or their interaction produced this traction, nor whether the narrative scope of a film affects its commercial success.

Across the sample window, all three component variables move upward. The MSI rises by 1.55 points, or 76% of its sample mean, while geographic scope shows the largest proportional increase at 120% of its mean. The components remain related but non-redundant: year-residualized pairwise correlations among *S*, *D* and *C* range from 0.57 to 0.68. The joint movement therefore suggests a coherent broadening of narrative scope rather than three projections of a single underlying score. All four trends are distinguishable from zero at conventional significance levels.

However, the rise is not smooth. Visually, Figure 11 suggests two periods of acceleration. The first period is in the early 1990s and the second one in the first half of the 2010s. These two sudden increases bracket longer plateaus, with a mild levelling-off in the most recent years. Whether these phases align with specific industry transitions is taken up in Section 7.2.

Figure 11: Yearly Means of Narrative-Scope Components (No MSI), 1977–2025



Notes: Annual means of victim level *S*, hero–victim distance *D*, and conflict scale *C*, each coded on a 0–3 scale and aggregated across the top 20 worldwide-gross films per year. Dashed lines show OLS trends. The MSI is omitted because it is mechanically constructed from *S* + *D*. Pairwise residual correlations among *S*, *D*, and *C* range from 0.57 to 0.68, indicating related but non-redundant dimensions. Source: own coding and estimation, detailed results in Table 3.

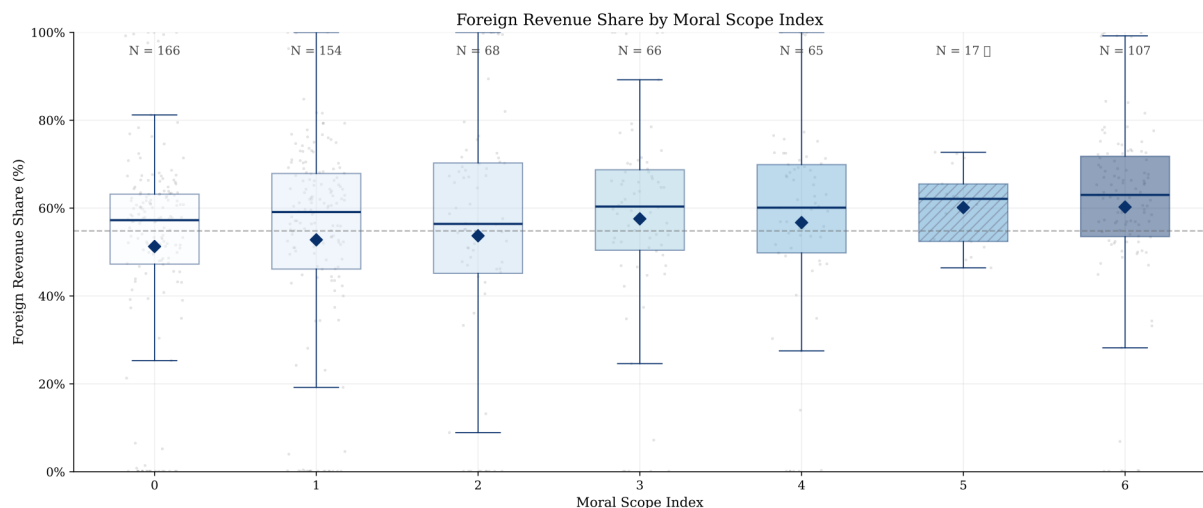
The decomposition of moral scope (Section 6.3) sharpens this reading. Of the rise in MSI, roughly 57% is carried by the victim-level component and 43% by hero-victim distance. Top-grossing films have moved somewhat further toward stories with broader victim categories than toward stories that place the hero at greater social distance from the people they protect. In conceptual terms, this is what a shift toward

broader moral concern in the sense of Enke et al. (2023) looks like when translated into a narrative setting: broader victim groups, greater social distance, and a wider geographic stage on which conflict unfolds. This finding complements existing cross-sectional evidence that morally resonant content matters for film success (Huang et al., 2025) by adding a temporal perspective on the narratives that succeed commercially. The magnitudes should not be overstated: even in 2025 the MSI mean sits just below 3 on a 0–6 scale, so the documented shift is a clear directional broadening, not a saturation of moral or geographic scope.

7.2 Alternative Explanations

The first acceleration phase, in the early 1990s, coincides with a period of expansion in the international theatrical market. As foreign revenue grew, the annual top 20 increasingly reflected global rather than primarily domestic demand. This transition is documented in recent surveys of the film-economics literature (McKenzie, 2023) and reflected in the rising foreign-revenue share of major releases (Motion Picture Association, 2022). A relationship between a film’s success internationally and broader moral and geographic stakes is plausible. Figure 12 shows that in the sample, films with higher MSI scores also have slightly higher foreign revenue shares (Spearman $\rho = 0.16$, $p < 0.01$). This pattern alone does not identify, an international-market channel.

Figure 12: Foreign Revenue Share by Moral Scope Index



Notes: Box plots with jittered observations show the distribution of foreign revenue share ($\text{foreign_pct} = \text{foreign_gross} / \text{worldwide_gross}$) across Moral Scope Index (MSI = S + D) scores. Sample: top 20 highest-grossing films worldwide per year, 1977–2025 ($n = 980$), restricted to films with both reported foreign revenue and a non-missing MSI ($N = 643$). Because Box Office Mojo’s international revenue coverage is incomplete before 1997, this restricted sample is effectively post-1997. Spearman rank correlation $\rho = 0.16$ ($p < 0.001$), indicating a small but positive association. whiskers extend to $1.5 \times \text{IQR}$, individual films are plotted as jittered points, per-score sample sizes indicated above each box. Source: authors coding with data from Box Office Mojo (domestic, foreign, and worldwide gross figures) retrieved on the 17.02.2026.

The second acceleration phase, in the first half of the 2010s, coincides with a rise of major-studio output around franchises, sequels, and shared-universe tentpoles. Such films reduce demand uncertainty (Dhar et al., 2012), concentrate marketing resources, and often rely on large-scale conflicts by design (Clement et al., 2014; Elberse, 2007). The saving-of-the-world architecture common in these films makes them more universal. Part of the documented rise in MSI may therefore reflect this compositional shift rather than a broadening of narrative scope across all commercially successful films. As before, the present design cannot separate the contribution of franchise consolidation from any underlying narrative trend.

The two mechanisms above are the most concrete alternative explanations but they are not exhaustive. The trend documented here is the joint outcome of many overlapping selection processes that the present design cannot distinguish. The same caution that governs the external interpretation of the trend applies equally to the design itself: how the indices were measured, where the sample begins and ends, and whether the documented direction of change can be cleanly distinguished from the feedback dynamics it likely participates in. The next section addresses those limits directly.

7.3 Limitations

The first concern is measurement. All four narrative scores are produced by a single LLM coder (GPT-5.4, see Chapter 4) applied uniformly across the 49-year sample. A common-mode bias would arise if that coder systematically rated later films higher than earlier films for reasons unrelated to the underlying material. For example, if its training skews toward recent narrative conventions. The year-residualized correlations among the three components offer a partial diagnostic. Pairwise film-level correlations move only marginally once the year trend is partialled out: S–D from 0.68 to 0.67, S–C from 0.61 to 0.58, D–C from 0.59 to 0.57 (Table D.3). Because residualization barely shifts these correlations, the co-movement among S, D, and C is structural rather than a shared-time-trend artefact, which is the pattern a year-driven common-mode bias would produce. The diagnostic is suggestive, not conclusive: a bias correlated with film characteristics rather than year would not be detected this way. A second, related simplification is the Moral Scope Index itself. Combining S and D into a single 0–6 score is deliberate but compresses two distinct dimensions, and the decomposition reported in Section 6.3 partially recovers what the aggregation hides.

The second concern is sample boundaries. International box-office figures from the early years of the window are less reliably reported than later ones. Before roughly 1990, foreign-revenue coverage in standard sources is incomplete, and worldwide-gross totals for early entries probably understate the foreign earnings of films from the late 1970s and 1980s. Some films from that period may therefore appear in the top 20 because their domestic performance dominated their reported total, when better data on foreign earnings would have replaced them with films of broader international reach. If those missing foreign earners systematically had broader moral and geographic scope, consistent with the cross-sectional pattern in Figure 12, then the early-window MSI mean is biased downward. In this case the documented rise overstates the true trend. The direction of this bias is the same as the bias one might expect from internationalization itself, and the descriptive design cannot separate them.

A second sample-related limit lies in the use of worldwide gross as the ranking criterion. Gross revenue is the product of ticket price and ticket sales, and ticket prices differ markedly across markets. A ticket sold in the United States or Western Europe contributes more to a film's worldwide gross than one sold in an economy with lower price levels. Films that draw large audiences in low-price markets are therefore underweighted relative to films that draw the same audiences in high-price markets, even though their reach in attendance terms is equivalent. The top 20 by worldwide gross is in this sense a measure of where commercial revenue concentrates, not of which films are most widely seen across the world. The trend documented here should be read as a trend in the kinds of films that succeed in markets weighted by relative ticket price.

The third concern is the conceptual framework itself. The hero–victim–villain relational coding scheme used in this thesis builds on the narrative economics literature and is not novel. However, in this analysis the actor-type taxonomy plays an auxiliary role. Typing heroes and villains is used mainly to attach the relational scales (S, D, C). It is thus not a direct application of an existing measurement instrument. The analytic weight in Chapters 6 and 7 is carried mainly by the relational scales. By applying these specific 0–3 scales and the choice to summarize two of them as a Moral Scope Index, the findings of this analysis can hardly be validated with the findings of existing literature. A different principled operationalization of the same underlying constructs would most likely yield different numerical magnitudes.

Especially the ties to morality as studied by Enke et al. (2023) need to be interpreted with caution. Enke and co-authors define moral universalism as "the extent to which people exhibit the same level of altruism and trust towards strangers as towards in-group members" (Enke et al., 2023, p. 1924). The definition has two components: altruism and trust. They elicit these through individual-level survey instruments. MSI also has two components, but they do not map cleanly onto Enke's pair. Hero-victim distance (D) tracks the altruism component well: it captures how socially distant the beneficiaries are from the hero. Victim level (S) measures the size of that group. Trust does not appear. MSI therefore captures only half of what Enke calls moral universalism, and it does so through narrative content rather than elicited preference. Although the two constructs are related, broader-scope narratives should be interpreted as a narrative analogue to moral universalism rather than as a direct measure of it.

8. Conclusion

This thesis investigated whether the narrative structure of commercially successful films has shifted over the past five decades, and if so, in what direction. The analysis connects research on the theatrical market with ideas and methods from the narrative-economics to study morality in film. To do so, this thesis coded the 20 highest-grossing films per year from 1977 to 2025 ($n = 980$) and analyzed them along two dimensions. Specifically, it asked whether their narratives have broadened in the moral concern embedded in the relations between hero, villain, and victim and in the geographic scale of the conflicts those narratives organize. The analysis is descriptive rather than causal.

Across 980 films and 49 years, the four narrative measures all moved in the same direction. The Moral Scope Index rose from a decade mean of about 1.5 in the late 1970s to about 3 in the 2010s, a gain of roughly 1.55 points or 76% of the sample mean. The geographic scale of conflict moved furthest in proportional terms, gaining about 0.94 points or 120% of its sample mean. All four trends were statistically distinguishable from zero at $p < 0.01$. The decomposition of moral scope showed that both components contributed: roughly 57% of the rise was carried by the victim-level component and 43% by hero-victim distance. Top-grossing films therefore moved somewhat further toward stories with broader victim categories than toward stories that placed the hero at greater social distance from the people they protected. The direction is consistent across measures and the magnitudes are substantial. However, even at its 2014 peak the Moral Scope Index sat around the middle of its 0–6 scale, and the trend appears to plateau in the 2020s.

The contribution of this thesis is threefold. First, it added a temporal dimension to a literature on film content that has been largely cross-sectional. Existing work has shown that morally resonant content correlates with success in a given year (Huang et al., 2025). This thesis shows how the moral structure of the most successful films has changed over five decades. Second, the analysis operationalized the hero–villain–victim relation as a set of measurable scales. The role-based view of narrative is well established in political and economic theory (Gehring & Grigoletto, 2025; Jones & McBeth, 2010), but it has rarely been turned into film-level numerical indices that can be tracked over time. Third, it provides a portable measurement instrument: a large-language-model coding pipeline applied uniformly across the 49-year sample produces internally consistent narrative measures from public metadata and subtitle

text. The pipeline can be extended to other corpora, such as streaming originals or television, if textual proxies are available.

Two extensions follow naturally from this work. First, the descriptive design here cannot identify whether the documented broadening of morality is due to audience preferences, studio strategy, the internationalization of the theatrical market, or other factors. Linking film-level narrative scores to release-country revenue shares, audience composition, or studio production decisions would begin to separate these channels. Second, the theatrical top 20 captures only one slice of the screen-content market. Extending the pipeline to streaming originals, traditional television or video games would test whether the broadening documented here is specific to theatrical market winners or part of a wider shift in commercial narrative structures and storytelling.

In modern economics, markets, narratives, beliefs, and behavior are increasingly studied as interconnected rather than separate phenomena. The theatrical market brings these elements together. While economics has paid considerable attention to this market, the narrative content of films has often been treated as a black box. Opening that box shifts the focus from films competing for attention to the narratives within them competing for attention. This matters because narratives are not neutral: they can reflect, reinforce, or challenge social values and interpretations. This thesis takes a step in that direction by offering a systematic, longitudinal mapping of moral scope among mainstream cinema's most commercially successful narratives, and by developing a portable measurement framework that future work can build on.

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10. Appendix

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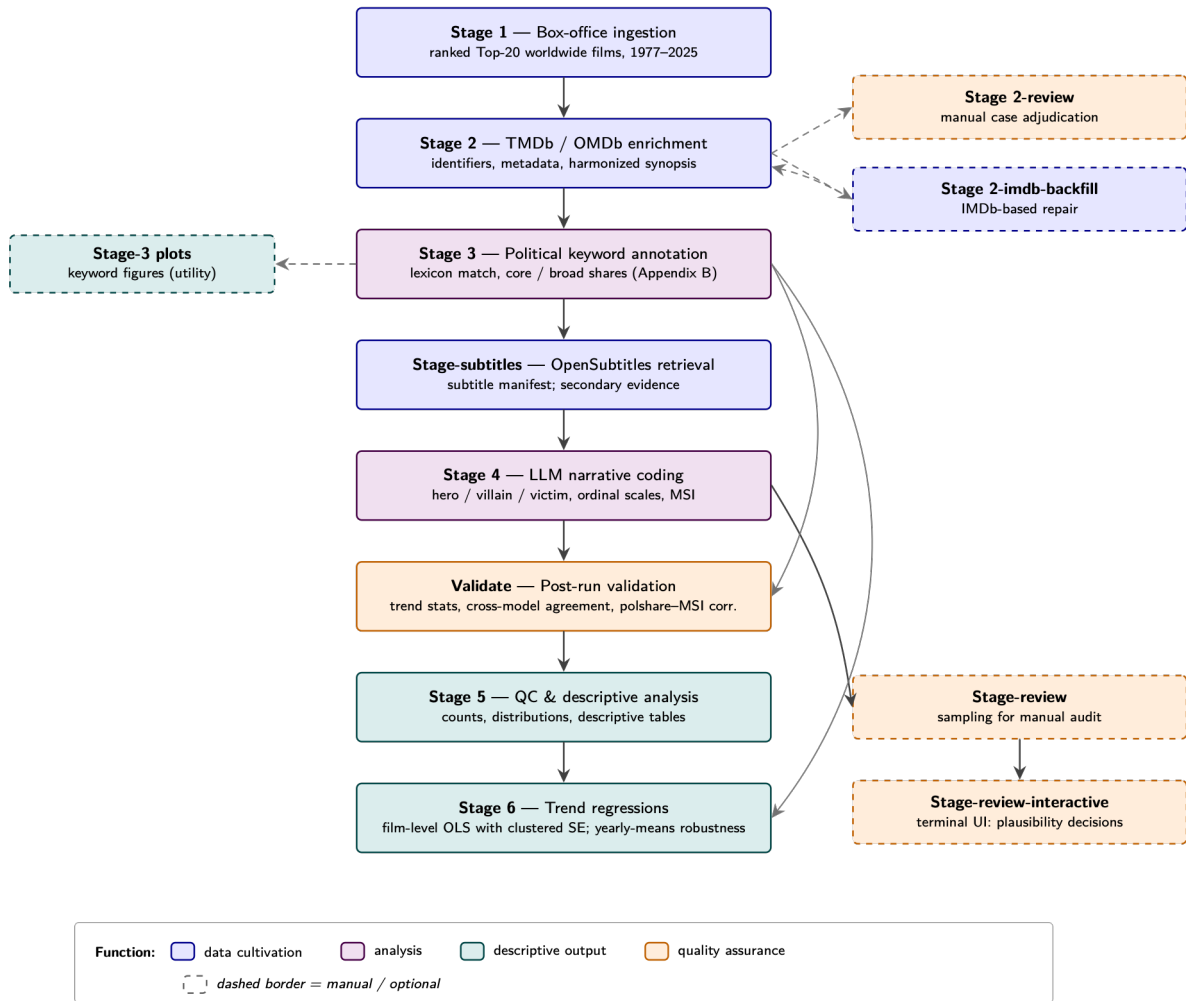
A. Narrative-Coding Pipeline

The full pipeline code, intermediate data files, and analysis scripts that produced the results in Chapter 6 are submitted alongside this thesis. Raw data from the external sources documented in Section 3.2 were accessed via their respective Application Programming Interfaces (APIs) and online interfaces between 03.08.2025 and 12.04.2026. Each external data point in the analysis dataset is traceable to its source through the identifier fields.

A.1 Architecture and Stage Overview

The Python project is organized as a staged command-line pipeline. Figure A.1 maps the full project. The pipeline is implemented as a sequence of modular stages organized into four functional groups: data-cultivation stages (blue) that build the film-level dataset, analytical stages (violet) that produce the substantive coding outputs, descriptive-output stages (teal) that summarize the coded data into tables, figures, and regression results, and quality-assurance stages (orange) that validate or audit the analytical outputs. The remainder of this section documents each stage in execution order.

Figure A.1: Architecture of the Boxoffice-Analysis Pipeline, Grouped by Functional Role.



Notes: Boxes denote individual pipeline stages, color-coded by functional role. Blue marks data-cultivation stages that build or extend the underlying film-level dataset (Stages 1, 2, Stage 2-imdb-backfill, Stage-subtitles); violet marks the two analytical stages that produce the substantive coding outputs (Stage 3 keyword annotation and Stage 4 LLM narrative coding); teal marks stages whose primary purpose is descriptive output (Stage-3 plots, Stage 5, Stage 6); orange marks quality-assurance stages that validate or audit the coded data (Stage 2-review, Validate, Stage-review, Stage-review-interactive). Solid borders mark stages that run as part of the automated run-all sequence; dashed borders mark manual or optional stages that are invoked separately. Solid arrows trace the main automated data flow; dashed arrows trace optional side branches and the IMDb-backfill loop into Stage 2. Stage 3 keyword outputs are also consumed by Validate and Stage 6 in addition to flowing forward through the main column.

The pipeline is implemented as a sequence of modular stages as shown in Figure A.1. Although not every stage is equally central to the thesis, each stage contributes to the construction, coding, or validation of the final analytical dataset. These stages are described in more detail below.

Stage 1 prepares the initial ranked box-office dataset. It serves as the ingestion stage of the project and creates the base film-level input on which later enrichment steps

operate. In practical terms, this stage transforms raw box-office information into a structured ranked dataset that can be passed to the metadata-enrichment workflow.

Stage 2 enriches the sample with external film identifiers and metadata. In this stage, films are matched to TMDb records, optionally enriched with OMDb information, and assigned a match-confidence status. Clear matches can be merged automatically, whereas uncertain cases are routed into a separate manual-review workflow. The same stage also harmonizes synopsis information, with OMDb plot text preferred where available and TMDb overview text used as fallback.

Stage 2-review is the manual branch for unresolved or uncertain metadata matches. It allows the user to inspect unresolved TMDb candidates interactively and either accept a proposed match, enter an IMDb identifier manually, or skip the case. This step is important because it prevents uncertain records from being merged automatically without inspection.

Stage 2-imdb-backfill complements the review step by using manually supplied IMDb identifiers to backfill missing synopsis or keyword information. This stage is therefore a repair and completion step for cases that could not be resolved automatically during the main Stage-2 run.

Stage 3 performs the political keyword analysis. It maps film-level keyword data to a political lexicon (Table B.1) and constructs keyword-based indicators, including distinctions between core and broad political content. The resulting variables are used for the descriptive benchmark analysis of political content in commercially successful films.

Stage 3 plotting utilities generate visualizations based on the Stage-3 outputs. These utilities are not part of the main coding pipeline itself, but they help produce descriptive keyword figures such as yearly keyword-share plots and alternative publication-style versions of the Stage-3 results.

Stage subtitles retrieves subtitle material and writes a subtitle manifest. Where subtitle files are found, they are downloaded and registered so that later stages can access them systematically. This stage supports the narrative-coding workflow, but subtitle material remains secondary to the synopsis in the final source hierarchy.

Stage 4 is the central analytical stage of the project. It combines the synopsis and optional subtitle evidence, extracts compact subtitle snippets where available, builds the final prompt inputs, sends the film-level material to the large language model (LLM), parses the JSON response, validates the returned fields, and writes the coded outputs. This stage also creates split output files, including a compact analysis-ready

file and a more detailed audit-oriented file. The prompt design used in this stage explicitly defines the synopsis as the primary source and subtitle evidence as secondary support.

Stage validate runs a set of post-coding checks against the Stage-4 outputs. It computes cross-model agreement statistics (Cohen's kappa and percentage agreement) between any two Stage-4 runs carried out with different language models, and the correlation between the Stage-3 keyword-based political-share indicator and the Stage-4 MSI. The results of these checks are written as separate CSV files in the validation output directory and underpin the cross-model robustness analysis reported in Section 4.4 and Appendix C.

Stage 5 performs data-quality checks and descriptive analysis. It verifies key structural properties of the analytical sample, such as expected counts per year and rank integrity, and produces descriptive tables and plots. This stage therefore functions as the project's main quality-control and output-generation layer after the coding run has been completed.

Stage 6 estimates the central trend regressions reported in Table 3. It operates on the analysis-ready Stage-4 output and, where relevant, on the Stage-3 keyword shares, fitting film-level OLS regressions of each ordinal narrative variable on a linear time index with standard errors clustered at the year level. The same stage also produces a yearly-means robustness table that re-estimates the same trends on year-aggregated data, providing a check that is insensitive to within-year variation in the size of the coded sample.

Stage review prepares the manual plausibility-review package. It selects all low-confidence cases and supplements them with a random subset of non-low-confidence cases. The resulting review files make it possible to inspect the coded output more systematically and to document the plausibility-review process separately from the automated coding pipeline. These files are the input to Stage-review-interactive.

Stage-review-interactive consumes the review file produced by Stage-review and presents the sampled cases one by one in a terminal interface for manual plausibility coding. Each case is displayed together with the model output, the underlying synopsis, and any subtitle evidence; the reviewer enters a plausibility judgment, an issue type if applicable, and any manual corrections, which are written back to the review CSV as separate columns. The decisions collected by this stage form the basis of the manual plausibility audit are reported in Section 4.4 and detailed in Appendix C.

A.2 Language Model Choice

The narrative coding produced in Chapter 4 depends on the behavior of the language model used as a coder. To mitigate the risk that the observed trends are artefacts of any particular model and its training, the thesis relies on two independently developed models from two different providers: GPT-5.4 (OpenAI) as the primary coder and Claude Opus 4.6 (Anthropic) as the validation coder. GPT-5.4 produces all results reported in Chapter 6. Claude Opus 4.6 is used only for the cross-model robustness check described in Section 4.4 and documented in full in Appendix C. The choice of two models from different providers and architectures strengthens the claim that the findings reflect signal in the material rather than properties of one model's training distribution or post-training alignment.

GPT-5.4 was selected as the primary coder for three reasons. First, the narrative schema defined in Section 4.2 requires constrained structured output: ordinal codes, fixed category labels, per-field reasoning, and explicit confidence levels. Recent GPT models handle this kind of schema-bound generation reliably and provide JSON output, which the pipeline relies on for parsing and validation (Chapter 4). Second, GPT-5.4 offers a sufficient context window to process each film's synopsis together with the evidence snippets extracted from its subtitles without truncation (Chapter 4). Third, using a widely available frontier model from one of the leading providers improves the reproducibility of the pipeline for future replications, since the exact model version and API surface are documented and publicly accessible.

All 980 films in the sample (top 20 worldwide box office per year, 1977–2025) were coded with GPT-5.4 under the settings recorded in the pipeline configuration (Appendix A). Output was constrained by the schema and prompts reproduced verbatim in Appendix A.3. Cases that could not be resolved within the schema were retained and labelled unresolved rather than imputed, following the policy described in Section 4.3.

Claude Opus 4.6 was selected as the validation coder because it represents the strongest frontier model of a second major provider (Anthropic) available at the time the validation run was carried out. Using a model from a different provider, trained on different data and aligned through different post-training procedures, makes the cross-model comparison a more demanding robustness test than re-running variants of the same model family.

The validation run used the identical sample ($n = 980$), the identical system and prompts (Appendix A.3), the identical subtitle-evidence preprocessing, and the identical output schema and parsing rules (Section 4.2). The only deliberate difference between the primary and validation run is the coding model itself. Results of the cross-model comparison are reported in Section 4.4 and detailed in Appendix C.

The two-model design shows that the trends over time in Chapter 6 are not just the product of one model's specific behavior. A second frontier model, trained independently by a different provider, produces the same direction, timing, and relative size of change across all moral-scope components (see Section 4.4 and Appendix C). This is not the same as human inter-coder reliability, because both coders rely on the same general technology (large language models). It also does not guarantee that earlier or future model versions would return identical coding decisions, since models change with each release. The design therefore serves as a cross-model robustness check that adds to, rather than replaces, the manual plausibility review documented in Section 4.4 and Appendix C.

A.3 Prompts

The Stage-4 narrative coding relies on two external prompt files: a system prompt and a user prompt template. Together, they define the coding task, the relevant source hierarchy, the output schema, and the film-specific inputs passed to the model. The system prompt contains the substantive coding rules. It specifies that the synopsis is the primary source, while subtitle evidence is secondary and should only be used to clarify or support the coding when helpful. It also defines the coding fields, the allowed values, the unresolved conventions, and the requirement that the model return valid JSON only with exactly the requested keys. The user prompt then inserts the film-specific materials into a fixed template using the title, year, synopsis, and subtitle evidence placeholder.

The prompts are reproduced below in full and without modification. They are included verbatim to document the exact instructions used in the final narrative-coding stage. This analysis does not treat the language model as a black box. Instead, the coding logic is made explicit through a fixed prompt structure, a restricted output contract, and a subsequent parser-validation step in Python.

A.3.1 System Prompt (verbatim)

You are a research assistant coding narrative variables for a thesis on film narratives. Analyze one film at a time using the provided materials.

Source priority:

- The synopsis is the primary source.
- Subtitle evidence is secondary.
- Use subtitle evidence only to clarify or support the coding when helpful.
- Do not let isolated subtitle lines override the overall narrative structure established by the synopsis.

Task:

Code the film's main narrative structure using these fields:

- hero_name
- hero_actor
- villain_name
- villain_actor
- victim_name
- victim_level
- hero_victim_distance
- conflict_scale
- confidence
- reasoning_summary

General rules:

- Code the main conflict, not a minor subplot.
- Focus on the central hero, central villain, main victim, and central conflict scale.
- Code the central stakes, not the largest imaginable stakes.
- Do not infer beyond the provided materials.
- Be conservative and consistent.
- If a value cannot be determined with reasonable confidence, use the unresolved option instead of forcing a guess.

Output Contract (strict):

- Return only valid JSON.
- Do not return markdown.
- Do not include any text outside the JSON.
- Do not output moral_scope_index.
- Do not output any extra keys.
- Return exactly these top-level keys:

hero_name, hero_actor, villain_name, villain_actor, victim_name, victim_level,
hero_victim_distance, conflict_scale, confidence, reasoning_summary

- Output raw labels only (never include definition text in values).
- Allowed values:

hero_actor, villain_actor -> individual_civilian | collective_civilian |
state_military | corporate_institutional | non_human | none

victim_level -> 0 | 1 | 2 | 3 | null

hero_victim_distance -> 0 | 1 | 2 | 3 | null

conflict_scale -> 0 | 1 | 2 | 3 | null

confidence -> high | medium | low

Variable Definitions:

hero_name:

- Main hero/protagonist entity in the central conflict; may be a named character or collective entity.

- If the entity cannot be identified, use null.

hero_actor:

- Actor type of the main protagonist responding to the central conflict.

- Label definitions:

individual_civilian = one non-state human actor (single protagonist)

collective_civilian = non-state collective (family, friends, team, community, crowd)

state_military = actor with sovereign/public authority or official state mandate (government, police, military, intelligence, state agency)

corporate_institutional = private or non-state formal organization (firm, nonprofit, school, hospital, media organization, church, etc.)

non_human = non-human entity (animal, monster, alien, AI, supernatural force, etc.)

none = actor type cannot be reasonably determined from available evidence

villain_name:

- Main villain/antagonist/threat entity in the central conflict; may be named or collective.
- If the entity cannot be identified, use null.

villain_actor:

- Actor type of the main antagonist or threat source in the central conflict.
- Label definitions:
 - individual_civilian = one non-state human antagonist
 - collective_civilian = non-state collective antagonist
 - state_military = state/public-authority antagonist in official role
 - corporate_institutional = private/non-state institutional antagonist
 - non_human = non-human antagonist/threat
 - none = actor type cannot be reasonably determined from available evidence

State vs corporate disambiguation rule:

- If the actor's role depends on sovereign public authority, code as state_military.
- If the actor's role depends on private/non-state organizational authority, code as corporate_institutional.
- In mixed cases, code by operative role in the central conflict.

victim_name:

- Main threatened victim entity in the central conflict (person, group, institution, country, humanity, world).
- If the entity cannot be identified, use null.

victim_level:

- Scale of the main threatened victim.
- Label definitions:
 - 0 = individual (one person)
 - 1 = small_group (family, crew, team, other limited group)
 - 2 = community_or_nation (town, city, broad community, country, national public)
 - 3 = humanity_or_planet (humanity, world, Earth, planetary life)

null = unresolved (insufficient evidence)

hero_victim_distance:

- Social distance between hero and main victim, relative to the hero.

- Label definitions:

0 = intimate (family, romantic partner, close friend)

1 = local_community (shared immediate local context)

2 = same_country_or_group (shared wider national/social-group identity)

3 = distant_or_universal (socially distant strangers, foreigners, or humanity at large)

null = unresolved (insufficient evidence)

conflict_scale:

- Geographic scale of the central conflict.

- Label definitions:

0 = local (household, neighborhood, school, town, city)

1 = national (primarily one country/national political space)

2 = international (cross-border or multi-country)

3 = global_or_planetary (world-level, humanity, planetary stakes)

null = unresolved (insufficient evidence)

Unresolved values:

- Use "none" for hero_actor or villain_actor only if actor type cannot be reasonably determined.

- Use null for victim_level, hero_victim_distance, or conflict_scale only if evidence is insufficient.

- Use null for hero_name, villain_name, or victim_name if entity cannot be identified.

Confidence:

- high = coding is clear and strongly supported

- medium = coding is plausible but requires interpretation

- low = materials are sparse, vague, or partially conflicting

Reasoning_summary:

Provide short justifications for:

- hero_actor
- villain_actor
- victim_level
- hero_victim_distance
- conflict_scale

Return JSON in this structure:

```
{
  "hero_name": "the protagonist",
  "hero_actor": "individual_civilian",
  "villain_name": "the main threat",
  "villain_actor": "collective_civilian",
  "victim_name": "the community",
  "victim_level": 2,
  "hero_victim_distance": 1,
  "conflict_scale": 1,
  "confidence": "medium",
  "reasoning_summary": {
    "hero_actor": "Main response is led by a single civilian protagonist.",
    "villain_actor": "Central threat comes from a civilian collective/group.",
    "victim_level": "Primary threatened victims are at community scale.",
    "hero_victim_distance": "Hero is closely tied to the same local community.",
    "conflict_scale": "Conflict is centered within one national setting."
  }
}
```

A.3.2 User Prompt Template (verbatim)

Film title: {title}

Year: {year}

SYNOPSIS (primary source):

{synopsis}

SUBTITLE EVIDENCE (secondary source; optional, may be empty):

{subtitle_text_placeholder}

Instruction:

Using the synopsis as the primary source and the subtitle evidence only as supporting context when helpful, code the film according to the schema and constraints specified in the system prompt. Output JSON only.

A.4 Inputs and Source Hierarchy

The Stage-4 coding procedure combines several input layers. At the most basic level, each row contains the identifying information required to code one film, including the film title and release year. To this, the pipeline adds harmonized synopsis text and, where available, subtitle-based evidence derived from subtitle files. In the analytical run relevant for this thesis, these inputs are attached to the annual top 20 sample and then passed one film at a time to the narrative-coding stage.

Subtitle files are processed to preserve spoken dialogue while removing technical and non-semantic noise. Preprocessing removes subtitle numbering, timestamps, markup tags, and hearing-impaired markers (e.g., music/laughter annotations), and normalizes whitespace. The cleaned text is then segmented into ordered dialogue blocks. Evidence extraction is rule-based and recall-oriented. Keyword groups (shown in Table A.1) linked to victim scope, social distance, and conflict scale are used to identify potentially relevant positions in the subtitle stream. Around these matches, context windows are extracted and overlapping windows are merged. The result is a compact set of context-rich snippets that can support disambiguation in narrative coding. Importantly, subtitle evidence is not treated as an independent classification source. It is used only as secondary context relative to the synopsis.

Table A.1: Subtitle Evidence Keyword Groups

Keyword Group	Purpose	Keywords
victim_scope	Retrieves dialogue segments that may indicate the scale of the threatened victim group.	daughter; son; wife; husband; family; child; children; friend; girl; boy; team; crew; group; villagers; city; town; country; nation; citizens; people; community; president; government; world; planet; humanity; mankind; everyone; earth; all of us
social_distance	Retrieves dialogue segments that may indicate the social distance between hero and victim.	my daughter; my son; my wife; my husband; my family; my friend; our town; our city; our people; our country; strangers; everyone; humanity; people out there; foreign; another country
conflict_scale	Retrieves dialogue segments that may indicate the geographic scale of the central conflict.	house; school; street; town; city; country; nation; president; government; army; countries; foreign; embassy; border; international; world; planet; earth; humanity; mankind; global

Notes: The table lists the keyword groups used to retrieve subtitle evidence snippets. Keywords are used for recall-oriented retrieval of contextual evidence only, they do not determine the final narrative codes. Final coding decisions are made by the LLM using the synopsis and retrieved subtitle evidence jointly. Source: author’s coding

The source hierarchy is intentionally conservative. Synopsis text serves as the primary basis for coding the main narrative structure of the film. Subtitle material is used only as secondary supporting evidence and is not intended to override the broader narrative logic established by the synopsis. This hierarchy is not only described in the thesis, but also directly encoded in the prompts. The system prompt explicitly states that the synopsis is the primary source and that subtitle evidence is secondary and should only be used when helpful for clarification or support. The same ordering appears in the user prompt, where the synopsis is presented first and subtitle evidence is listed separately as optional secondary material.

In practical terms, this means that the coding stage can proceed without subtitle evidence, but not without a usable synopsis. Rows with missing synopsis text are therefore not force-coded. Where subtitle material is available, it is first cleaned and reduced to compact evidence snippets before being inserted into the prompt. The purpose of this step is not to replace the synopsis, but to provide additional context in cases where the synopsis is short, ambiguous, or in need of support.

This arrangement reflects the substantive goals of the project. The synopsis provides a relatively standardized and comparable representation of the central plot

across films, whereas subtitle files can supply more detailed but also noisier contextual information. By design, the coding procedure therefore treats subtitle evidence as subordinate to the synopsis rather than as an equal or competing source.

A.5 Outputs

The Stage-4 coding stage produces both substantive narrative variables and process-oriented audit information. The substantive outputs correspond directly to the coding framework developed in the main chapters of the thesis. These include the identified hero, villain, and victim entities, the actor-type codings for hero and villain, the three ordinal scale variables (`victim_level`, `hero_victim_distance`, and `conflict_scale`), a confidence label, and a short `reasoning_summary` object. According to the system prompt, the model is required to return exactly these top-level keys and no additional fields. Table A.2 shows the allowed values of the output schema, including the data type for each field.

Table A.2: Output Variables of the Stage-4 Narrative Coding.

Variable	Data type	Allowed values
<code>hero_name</code>	string null	free text or null
<code>villain_name</code>	string null	free text or null
<code>victim_name</code>	string null	free text or null
<code>hero_actor</code> , <code>villain_actor</code>	string	individual_civilian, collective_civilian, state_military, corporate_institutional, non_human, none
<code>victim_level</code>	integer null	0, 1, 2, 3, null
<code>hero_victim_distance</code>	integer null	0, 1, 2, 3, null
<code>conflict_scale</code>	integer null	0, 1, 2, 3, null
<code>confidence</code>	string	high, medium, low
<code>reasoning_summary</code>	object	exactly 5 subkeys

Notes: Each row defines one top-level field that the language model is required to return for every film. String fields ending in `"_name"` identify the relevant entity in free text; null is used when the entity cannot be identified from the available materials. The actor fields are restricted to a closed set of categories defined in Section 4.2, with "none" reserved for cases where the actor type cannot be reasonably determined. The three ordinal scale variables are restricted to integers 0–3, with null reserved for cases where evidence is insufficient (Section 4.3). `Reasoning_summary` is a JSON object with exactly five subkeys, one per coded variable (`hero_actor`, `villain_actor`, `victim_level`, `hero_victim_distance`, `conflict_scale`). The system prompt enforces that no other top-level keys are returned. Source: own pipeline, full prompt reproduced verbatim in A.3.

The returned values are tightly constrained. The actor fields `hero_actor` and `villain_actor` are restricted to a predefined set of actor categories including the category none when the actor type cannot be determined with reasonable confidence. The scale variables are restricted to the values 0, 1, 2, 3, or null if the available evidence is insufficient. Confidence is restricted to the

values high, medium, and low. This restricted output design makes the coded dataset easier to validate, parse, and compare across films.

A central methodological point is that the MSI is not generated by the language model itself. The system prompt explicitly instructs the model not to output this variable. Instead, the MSI is constructed deterministically in the Python pipeline after parsing, based on the coded values of S and D. This is useful because it separates model-generated coding from rule-based variable construction.

In addition to the substantive coding outputs, the pipeline stores process and audit fields. These document whether a row was parsed successfully, whether warnings occurred, whether the row had to be skipped, and what raw or intermediate materials were associated with the coding call. The project therefore produces not only a compact analysis-ready dataset, but also more detailed audit-oriented outputs that preserve row-level trace information for later inspection.

Unresolved cases are handled explicitly rather than hidden. The prompt instructs the model to use none when an actor type cannot be reasonably determined and null when entity names or scale values cannot be identified with sufficient confidence. In this way, uncertain cases are retained as informative measurement outcomes rather than being forced into substantive categories. This unresolved logic is important for the interpretation of the final dataset, because it ensures that ambiguity remains visible in the output rather than being concealed through artificial precision.

A.6 Quickstart

The public GitHub repository contains the complete Python pipeline code, configuration templates, prompts, validation utilities, tests, and documentation required to understand and reproduce the project workflow. It is structured to document all analysis stages transparently, including narrative coding, quality checks, visualization, and robustness comparison logic. For practical and legal reasons, large runtime datasets and raw subtitle files are not included in the public repository, and API credentials are removed entirely. The repository therefore serves as the canonical, auditable code base, while data-dependent outputs are generated locally from configured inputs. The full project, including the complete working directory and accompanying materials, is submitted separately as a ZIP archive as part of the full thesis package.

GitHub repository: <https://github.com/Cedric-Roetheli/boxoffice-analysis-pipeline>

B. Keyword Analysis

This appendix documents the keyword-based analysis of political content. Unlike the LLM-based narrative coding (Chapter 4, Appendix A and C), this module uses a curated lexicon to flag political topicality in film-level keyword metadata. Its purpose is not to recover narrative structure but to provide a transparent, fully reproducible benchmark for the presence of political content in commercially successful films. The remainder of this appendix focuses on the construction of the measure: the lexicon itself (Table B.1), the normalization and matching rules, and the variables that the stage produces. Period-level descriptives for these variables are reported in Table B.3 and the accompanying figures of Appendix B.5.

B.1 Keyword-Analysis Pipeline

The keyword analysis is implemented as Stage 3 of the boxoffice-analysis pipeline. The pipeline architecture as a whole, including how Stage 3 is positioned between metadata enrichment (Stage 2) and downstream coding (Stage 4) and analysis (Stages 5 and 6), is documented in Appendix A.1 and is not repeated here. The relevant interface for this appendix is narrow: Stage 3 reads the film-level field `tmdb_keywords` produced by Stage 2 and writes a set of keyword-based indicators and shares back into the master film table. The companion Stage-3 plotting utilities (also documented in Appendix A.1) draw on those outputs for the figures shown in Appendix D.2 but are otherwise unrelated to the matching procedure documented below.

B.2 Political Lexicon

The active lexicon used by the project is `political_lexicon_v1.csv`. It contains 421 terms organized as a four-column table: `term`, `level1`, `level2`, and `include_core`. The `level1` column places each term in one of four top-level categories (policy, politics, polity, peripheral political context) and `level2` records a finer sub-category within each. The `include_core` flag determines whether a term is counted toward the strict core measure (`include_core` = 1) or only toward the inclusive broad measure (`include_core` = 0). In `political_lexicon_v1.csv` the three substantively political top-level categories (policy, politics, polity) are entirely core, and peripheral political context is entirely broad-only. The broad measure is therefore strictly nested over the core measure. Every core match is also a broad match, but not the other way round.

Table B.1 gives the full view of the lexicon. For each top-level category it lists the `level2` sub-categories it contains and reports the term list separately for each of the 24 `level1` × `level2` combinations. The tables makes the entire lexicon inspectable in one page. The lexicon is curated by hand rather than learned from data. It is best understood as a transparent rule-based classification device that translates film-level keyword metadata into interpretable political-content indicators, and it is versioned.

The substantive logic behind the core / broad split is the following. The core measure aims at clearly political content (the institutions, offices, decisions, contests, and policies of organised political life) and is populated by the policy, politics, and polity categories. The broad measure adds a layer of peripheral political context: terms that frequently appear in the immediate vicinity of political content (army, police, corruption-adjacent class markers, public-issue plot markers) but that, taken on their own, are too generic to count as political content in a strict sense. A keyword can therefore register on the broad measure without registering on the core measure. The opposite is impossible by construction.

Table B.1: Political Lexicon

	(1)	(2)	(3)
Top-level category	Level2 sub-category	Terms	n
policy	border migration	asylum; asylum_seeker; border_control; border_patrol; border_security; citizenship; deportation; detention_center; immigration; immigration_policy; internment_camp; migrant; migration; naturalization; refugee; undocumented_immigrant; visa_policy	17
	crime corruption regulation	antitrust; big_business; bribe; bribery; cartel; corporate_crime; corporate_power; corruption; embezzlement; fraud; money_laundering; monopoly; oligarch; oligarchy; regulatory_capture; scandal; white_collar_crime	17
	economy finance	austerity; bailout; bank; banking; budget; central_bank; credit; debt; deficit; depression; development_policy; economic_policy; economics; economy; finance; financial_crisis; fiscal_policy; foreclosure; inflation; loan; monetary_policy; mortgage; nationalization; privatization; public_spending; recession; stock_market; subsidy; tariff; taxation; taxes; trade_policy; wall_street	33
	foreign affairs diplomacy	alliance; arms_control; ceasefire; diplomacy; diplomatic_mission; disarmament; foreign_policy; international_relations; negotiation; peace_treaty; peacekeeping; sanctions; summit_meeting; treaty	14
	rights equality order	abortion_law; apartheid; censorship; civil_rights; discrimination; equal_rights; freedom_of_press; freedom_of_speech; human_rights; indigenous_rights; minority_rights; political_persecution; racism; religious_freedom; reproductive_rights; same_sex_marriage; segregation; voting_rights; womens_rights	19
	security surveillance	anti_terrorism; counterinsurgency; counterintelligence; counterterrorism; domestic_spying; emergency_powers; intelligence_operation; national_security; security_policy; surveillance; terrorism; terrorist; wiretap	13
	war conflict	armed_conflict; civil_war; cold_war; coup; coup_detat; guerrilla; insurgency; insurgent; insurrection; invasion; military_intervention; occupation; occupation_force; proxy_war; rebellion; regime_change; resistance_movement; revolution; uprising; war; world_war_i; world_war_ii	22
	welfare social policy	education_policy; food_policy; healthcare_policy; housing_policy; labor_policy; minimum_wage; pension_reform; poverty_reduction; public_health; redistribution; social_policy; social_security; unemployment; welfare; welfare_state	15
politics	decision power struggle	backroom_deal; cabinet_crisis; filibuster; government_crisis; impeachment; legislative_bargaining; lobbying; lobbyist; no_confidence_vote; parliamentary_crisis; political_negotiation; power_struggle; resignation; succession_struggle; veto	15
	elections representation	ballot; ballot_box; campaign; campaign_manager; election; election_campaign; elections; local_election; parliamentary_election; plebiscite; political_campaign; polling_station; presidential_election; primary_election; recount; reelection; referendum; runoff_election; vote; votes; voting	21
	ideology regime conflict	authoritarian; authoritarianism; colonialism; communism; communist; democracy; democratic; dictator; dictatorship; fascism; fascist; ideological_conflict; ideology; imperialism; nationalism; nationalist; one_party_state; political; politics; populism; populist; propaganda; revolutionary_movement; socialism; socialist; totalitarianism	26
	participation representation	civic_engagement; constituency; political_participation; popular_mobilization; public_opinion; representation; voter_turnout	7
	parties competition	bipartisanship; coalition; coalition_government; leadership_contest; opposition; parliamentary_debate; partisanship; party_congress; party_leader; party_membership; party_politics; political_machine; political_party; ruling_party	14
	protest mobilization	activism; activist; boycott; civil_disobedience; demonstration; general_strike; grassroots_campaign; labor_movement; march; movement; occupation_protest; protest; protest_movement; rally; riot; sit_in; social_movement; strike; student_movement	19
polity	constitution law	constitution; constitutional; constitutional_court; constitutional_law; court; decree; emergency_rule; executive_order; federal_court; high_court; judge; judicial; judiciary; justice; law; legal; legal_system; legislation; legislative; martial_law; ordinance; state_of_emergency; statute; supreme_court	24
	institutions government	administration; assembly; cabinet; chancellor; congress; executive; governance; government; house_of_representatives; legislature; lower_house; minister; ministry; national_assembly; parliament; presidency; prime_minister; public_administration; regime; senate; state; upper_house	22
	offices roles	ambassador; bureaucrat; civil_servant; congressman; congresswoman; consul; diplomat; governor; legislator; mayor; member_of_parliament; mp; president; public_official; senator; statesman; vice_president	17
	security state apparatus	border_guard; cia; customs; fbi; homeland_security; intelligence; intelligence_agency; kgb; military_government; police_state; secret_police; secret_service; security_service; state_security; surveillance_state	15
	territory sovereignty	annexation; border; border_crossing; border_zone; colonial_rule; colony; consulate; decolonization; embassy; frontier; occupation_zone; occupied_territory; protectorate; sovereign_state; sovereignty; territory	16
peripheral political context	broad public issue plotmarkers	civic_life; crisis; institution; institutions; policy; public_affairs; public_policy; reform; reforms; social_order	10
	class economy plotmarkers	billionaire; capitalism; capitalist; class_conflict; class_warfare; corporate; corporation; greed; inequality; labor; labour; poverty; social_class; the_rich; union; worker; workers; working_class	18
	conflict hierarchy plotmarkers	authorities; authority; bureaucracy; crackdown; disorder; elite; elites; establishment; hierarchy; leader; ruler; rulers; ruling_elite; strongman; unrest	15
	law order plotmarkers	arrest; cop; criminal_justice; detention; incarceration; jail; law_enforcement; police; policing; prison; prison_guard; prisoner; trial	13
	war security plotmarkers	army; battle; bioweapon; chemical_weapon; combat; covert_operation; espionage; hostage; mercenary; military; missile; nuclear; secret_mission; soldier; soldiers; spy; veteran; veterans; war_zone	19
Total	24 sub-categories		421

Notes: Reproduces the structure of political_lexicon_v1.csv. Term forms are lowercase, underscore-joined to match the normalization of TMDb keyword metadata, matching procedure in B.3.

B.3 Normalization and Matching

Before matching takes place, both the film keywords and the lexicon terms are normalized. The strings are lowercased, separators (spaces, underscores, hyphens) are harmonized into a single underscore-joined form, and multiword expressions are preserved as connected units rather than split into independent words. This is important because many politically relevant terms are inherently multiword (e.g., `world_war_ii`, `freedom_of_speech`, `border_control`) and would lose their meaning if tokenized apart.

After normalization, each token in a multiword phrase is stemmed and the stemmed tokens are recombined into a phrase-level stem key. Matching is then performed by exact equality on this stemmed representation, so that surface-form variants such as `finance / finances`, `soldier / soldiers`, or `worker / workers` collapse onto a single lexicon entry. The procedure deliberately does not use semantic similarity or fuzzy matching. A TMDb keyword either has the same stem key as a lexicon term, or it is not recorded as political. The implementation also supports two film-level deduplication modes: `normalized` (deduplicate at the level of the normalised string) and `stem` (deduplicate at the level of the stem key). This affects only the denominator used in the keyword-share measures, not the matching itself. All results reported in this thesis use stem mode.

Table B.2 shows how the matching plays out in five films chosen to span the substantively interesting cells of the core $\times$ broad cross-tab: a high-core, high-broad case (e.g., *The Big Short*), a moderate-core case in which the broad measure picks up the surrounding war-genre vocabulary (e.g., *Full Metal Jacket*), a paradigmatic low-core case in which the broad measure adds nothing (e.g., *Star Wars: Episode IV*), a pure broad-only match (e.g., *Smokey and the Bandit*), and a film with no political signal on either measure (e.g., *Titanic*).

Table B.2: Worked Examples of Keyword Matching in Five Films.

Film	(1) Year	(2) Total kw	(3) Core (n / share)	(4) Broad (n / share)	(5) Core matches	(6) Broad-only matches
The Big Short	2015	14	7 / 0.500	7 / 0.500	bank, finances, financial_crisis, fraud, loan, mortgage, wall_street	—
Full Metal Jacket	1987	36	3 / 0.083	7 / 0.194	propaganda, racism, war	army, combat, military, soldiers
Star Wars: Episode IV - A New Hope	1977	22	2 / 0.091	2 / 0.091	rebellion, totalitarianism	—
Smokey and the Bandit	1977	23	0 / 0.000	1 / 0.043	—	cop
Titanic	1997	22	0 / 0.000	0 / 0.000	—	—

Notes: Five films chosen to illustrate the four substantively distinct cells of the core x broad cross-tab. polkw_core counts matches against the strict core lexicon (policy + politics + polity), polkw_broad counts matches against the inclusive broad lexicon (core + peripheral political context). polshare_core and polshare_broad rescale these counts by the film's total TMDb keyword count (total_kw). Matched terms are reported in their lexicon (stemmed and underscore-joined) form, broad-only terms are the matches that contribute to broad but not to core.

B.4 Output Variables

Stage 3 writes both substantive output variables and more detailed audit-oriented mapping fields back to the master film table. The main substantive outputs are the binary indicators political_any_core and political_any_broad, the matched-keyword counts polkw_core and polkw_broad, the keyword denominator total_kw, and the corresponding share measures polshare_core and polshare_broad. At the film level, these variables answer three questions. (i) Does the film register any political content (extensive margin)? (ii) How many of its TMDb keywords match the lexicon? (iii) What share of its keyword set is political (intensive margin)? Period-level summaries of political_any_* and polshare_* for both the top-20 and top-100 samples are reported in Table B.3 the accompanying yearly-trend figures appear in Appendix B.5.

Calculation of the Keyword Analysis Measures

The unit of observation at this stage is the film. For each film i , K_i denotes the set of TMDb keyword tags attached to that film. The normalized keywords are then matched against the political lexicon described in Section B.2. The lexicon distinguishes between two versions of political content. The broad version includes all keywords matched to the political lexicon. The core version includes only those matched keywords that are marked with *include_core* = 1 in the lexicon file. The main appendix figures use the broad version, while the core version is retained as a stricter alternative. Two measures of political content are computed for each film.

(i) The extensive margin records whether a film contains at least one political keyword. It is defined as a binary indicator:

$$political_any_i = 1 \text{ if } polkw_i \geq 1; 0 \text{ if } polkw_i = 0 \text{ (B1)}$$

where $polkw_i$ is the number of political keywords matched for film i . In the broad version, this corresponds to $polkw_broad$. In the core version, it corresponds to $polkw_core$.

(ii) The intensive margin measures how large the political keyword share is within a film's keyword set. It is calculated as:

$$polshare_i = polkw_i / total_kw_i \text{ (B2)}$$

where $total_kw_i$ is the total number of TMDb keywords available for film i . If a film has no keyword tags ($total_kw_i = 0$), the political keyword share is set to zero.

The same logic is applied separately to the broad and core lexicon versions:

$$polshare_broad_i = polkw_broad_i / total_kw_i \text{ (B3)}$$

$$polshare_core_i = polkw_core_i / total_kw_i \text{ (B4)}$$

At the year level, the extensive margin is computed as the share of films in a given year with at least one political keyword:

$$share_with_polkw_t = mean(political_any_i) \text{ (B5)}$$

The intensive margin is computed as the average political keyword share across films in the same year:

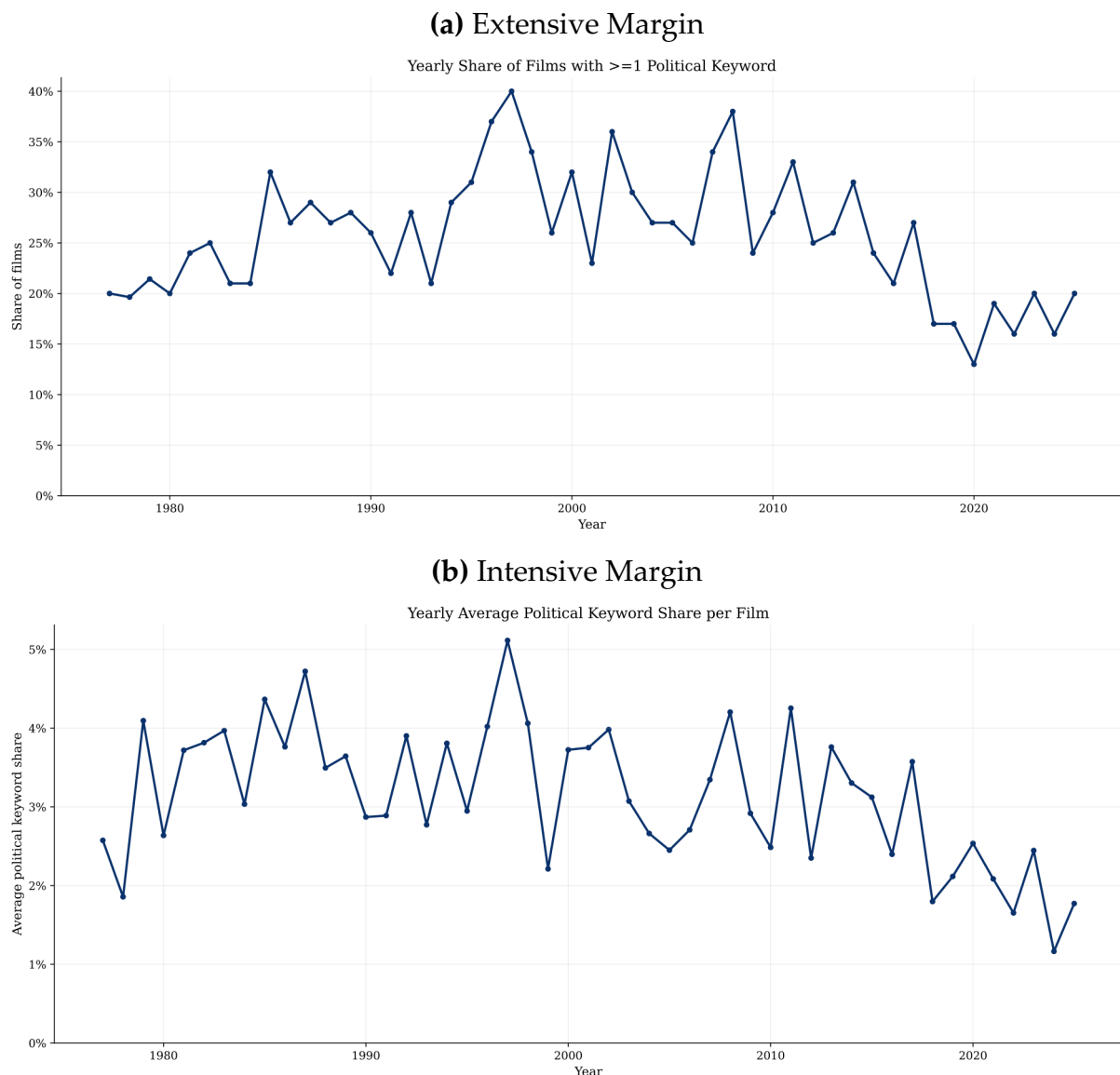
$$avg_polshare_per_film_t = mean(polshare_i) \text{ (B6)}$$

This film-level averaging gives each film the same weight within a year.

B.5 Results

This section reports the results of the keyword analysis described in Appendix B.1–B.4. The results are shown for the auxiliary top 100 sample from 1977 to 2025, because this broader sample provides a descriptive benchmark for the political visibility of commercially successful films beyond the main top 20 sample. Figure B.1 reports two yearly measures based on the broad political lexicon: the extensive margin and the intensive margin (defined in B.4).

Figure B.1: Political Visibility in Top-Grossing Films, 1977–2025.



Notes: Panel (a) shows the yearly share of films registering at least one match against the broad political lexicon. Panel (b) shows the yearly mean of the within-film political keyword share, defined as the number of broad-lexicon matches divided by the film’s total TMDb keyword count. The extensive margin captures whether political content is present at all, while the intensive margin captures how large the political keyword share is within a film’s keyword set. Sample: 100 highest-grossing films worldwide per calendar year, 1977–2025 ($n = 4,780$, coverage is partial in 1977 (40 films), 1978 (56), and 1979 (84), and complete from 1980 onward). Source: own calculation from TMDb keyword metadata.

Table B.3 summarizes the same measures by period and compares the broad and core lexicons across both the top 100 and top 20 samples.

Table B.3: Political-Keyword Presence in Commercially Successful Films, by Decade, 1977–2025

	(1)	(2)	(3)	(4)	(5)
	n	Ext. (core)	Ext. (broad)	Int. (core)	Int. (broad)
<i>Panel A: top 100 sample (auxiliary keyword sample)</i>					
1977–1979	180	13.3%	20.6%	1.9%	3.1%
1980–1989	1,000	16.6%	25.4%	2.1%	3.7%
1990–1999	1,000	19.1%	29.4%	1.9%	3.5%
2000–2009	1,000	21.3%	29.6%	2.1%	3.3%
2010–2019	1,000	16.8%	24.9%	1.8%	2.9%
2020–2025	600	11.7%	17.3%	1.0%	1.9%
All years	4,780	17.4%	25.8%	1.9%	3.2%
<i>Panel B: top 20 sample (main thesis sample)</i>					
1977–1979	60	11.7%	20.0%	0.7%	1.9%
1980–1989	200	21.0%	31.0%	1.9%	3.2%
1990–1999	200	30.5%	40.5%	2.5%	4.0%
2000–2009	200	22.5%	32.5%	1.6%	2.7%
2010–2019	200	15.5%	26.0%	1.1%	2.5%
2020–2025	120	13.3%	21.7%	1.2%	2.1%
All years	980	20.6%	30.4%	1.6%	2.9%

Notes: Each cell reports the period mean across all films in the indicated sample. Ext. (extensive margin) is the share of films containing at least one political keyword. Int. (intensive margin) is the mean share of a film's TMDb keywords classified as political. Core applies the strict political lexicon, broad applies the inclusive political lexicon (which nests core, see Appendix B). The top 100 sample is partly populated in 1977 (40 films), 1978 (56) and 1979 (84) and complete from 1980 onward, the top 20 sample contains exactly 20 films per year for all 49 years (n = 980). The 1977–1979 and 2020–2025 rows cover three and six years respectively, all other rows cover full decades. Source: own calculation from TMDb keyword metadata, lexicon and matching procedure documented in Appendix B.

Across both measures and both lexicons, political content is a recurring feature of commercially successful films throughout the sample period. The extensive margin rises from the late 1970s through the 1990s and 2000s before declining in the 2010s and

more sharply in the 2020s. The intensive margin moves on a smaller scale but follows a similar arc. Importantly these measures capture the topical presence of politics, not the narrative structure of the films, which is the subject of the main analysis of this Thesis.

C. Plausibility Review and Validation

This appendix documents the two validation layers summarized in Section 4.4: the manual plausibility review of a random subsample and the full cross-model replication of the coding run. It also records the automated validation steps that run inside the pipeline. The coded dataset used in Chapter 6 is produced by a large language model (GPT-5.4) and is therefore validated on several layers. The goal of this appendix is to give an auditable account of how the coding was checked. Three layers are reported: (i) automated validation built into the pipeline, (ii) a manual plausibility review on a random subsample, and (iii) a full cross-model replication of the coding on the same sample.

C.1 Automated Validation in the Pipeline

The pipeline performs two automated checks before a film enters the analysis dataset. The first is a schema-level parsing check: every model output is parsed against the narrative schema defined in Section 4.2, and any field that cannot be resolved to an allowed value is flagged and labelled unresolved (Section 4.3). The second is a quality-control layer that screens for implausible combinations of fields and malformed outputs, again labelling such cases unresolved rather than discarding them. The effect of both checks is to keep every film in the sample while making coding uncertainty visible, so that the downstream analysis can decide how to treat such cases.

C.2 Plausibility Review Results

The manual plausibility review is based on a 10% random subsample of the full sample of 980 films ($n = 98$). The subsample is stratified by year so that each of the 49 years in the sample period contributes a proportional share of cases, and it is drawn with a fixed random seed (42) to make the draw reproducible. The confidence distribution of the subsample is 32 high-confidence cases (33%), 57 medium-confidence cases (58%), and 9 low-confidence cases (9%), which is consistent with the confidence distribution of the full sample. Each of the 98 sampled cases was reviewed interactively. The

reviewer saw the film's metadata, its synopsis, the full model output (hero, villain, victim role, ordinal codes, per-field reasoning, and the model's own confidence), and the parsing status. Every decision was saved to the review log.

Of the 98 reviewed cases, 84 were judged plausible (85.7%) and 14 were judged implausible (14.3%). Implausibility is not spread evenly across the model's own confidence levels, as shown in Table C.1.

Table C.1: Plausibility Rate by Declared Model Confidence (n = 98).

Confidence	N Reviewed	Plausible	Implausible	Implausibility Rate
High	32	32	0	0.0%
Medium	57	47	10	17.5%
Low	9	5	4	44.4%
Total	98	84	14	14.3%

Notes: Each row reports the manual-review outcome for cases at the indicated level of declared model confidence. The confidence level is the model's own self-reported confidence flag (high, medium, low) attached to each coding output. Plausible and implausible are the reviewer's binary judgement on whether the full coded record (hero, villain, victim role, ordinal codes, and per-field reasoning) is consistent with the film's metadata and synopsis. Implausibility rate is the share of reviewed cases in each row judged implausible. Sample: 10% stratified random subsample of the 980-film top-20 sample, drawn with seed 42 (n = 98). Source: own review log.

The difference across confidence levels has two consequences for the analysis. First, it shows that the model's own confidence flag is a real signal. Cases declared as high-confidence are almost never judged implausible on manual review, while low-confidence cases are judged implausible about four times more often than the sample average. Second, it supports the rule introduced in Section 4.3 of keeping low-confidence cases as unresolved in the analysis, rather than forcing a decision, since these are the cases where the coding is most likely to be incorrect. Implausible cases are preserved in the review log but are not individually re-coded, consistent with the stated purpose of the review as a quality check rather than a full re-coding exercise.

The full review log, including the per-case decisions and session report, is stored in the python project under runs/stage_review/.

C.3 Cross-Model Validation

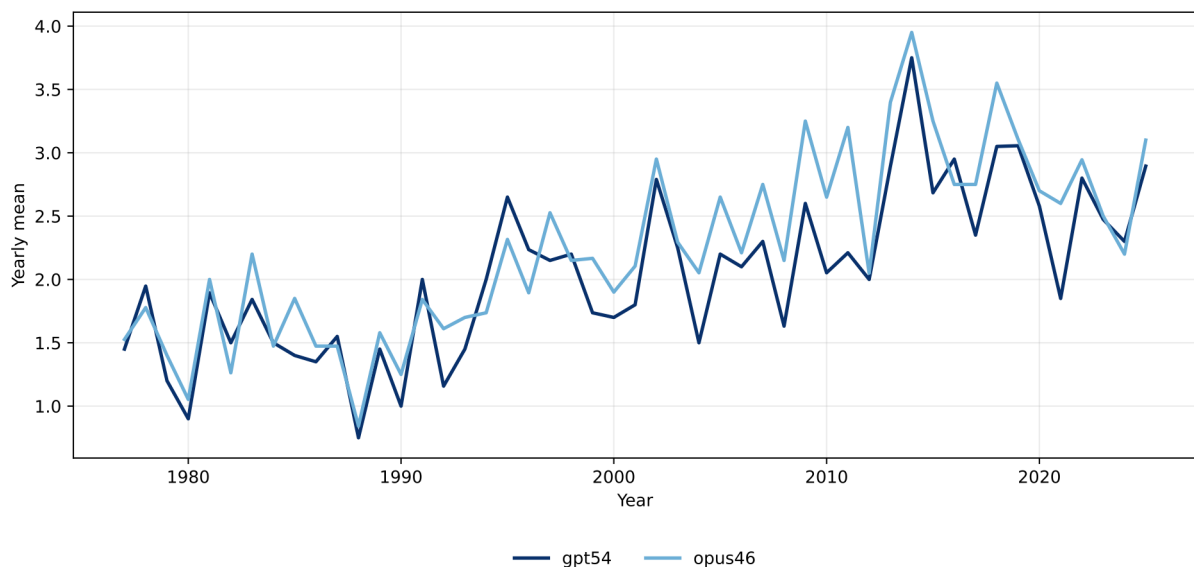
The cross-model validation replicates the full coding run on the identical sample of 980 films with Claude Opus 4.6 (Anthropic), a second frontier model from a different provider. The primary coder is GPT-5.4 (OpenAI). The choice of models and the reason for using two providers are documented in Appendix A.2. Both runs used identical prompts (Appendix A.3), identical subtitle-evidence preprocessing (Section 4.1), and identical output schema and parsing rules (Section 4.2). The only intended difference between the two runs is the coding model itself.

Item-level agreement between the two runs is reported in Table 2 in Section 4.4 and is not repeated here. This appendix focuses on the trend-level comparison, which is the question that matters for the analysis: does the temporal pattern depend on the choice of coder?

Trend-Level Comparison

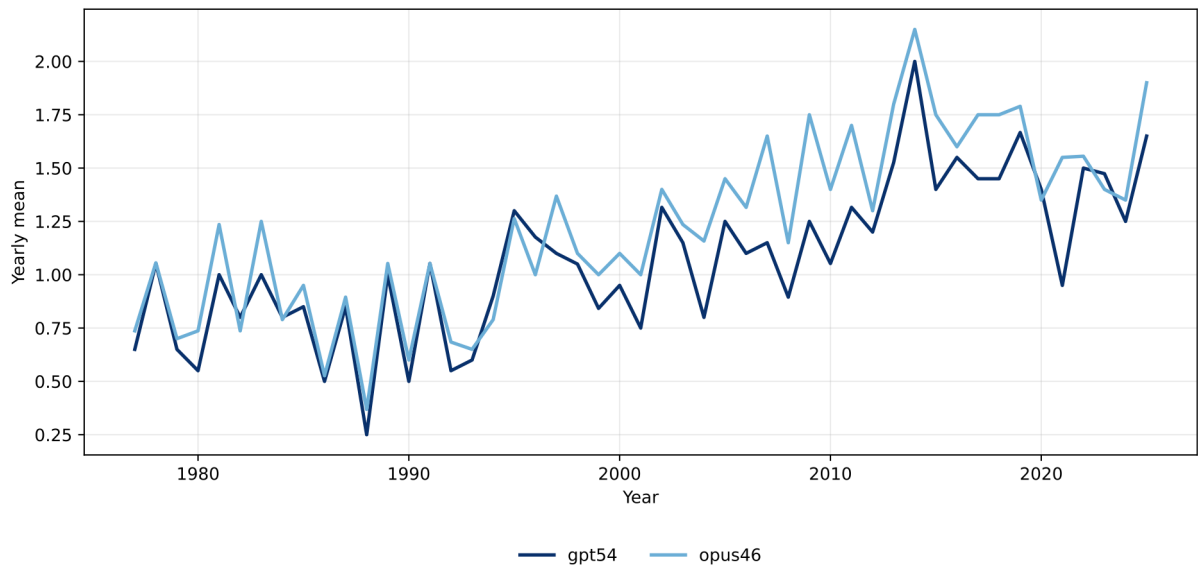
Figures C.1 to C.4 show the year-by-year means of the two runs for the composite MSI and for each of the three relational variables. The two series broadly track each other in direction and timing across all four variables. Claude Opus 4.6 sits slightly above GPT-5.4 in most years, so the absolute level of the coding is somewhat dependent on the model. The shape of the trend is not.

Figure C.1: Mean MSI Over Time: GPT-5.4 vs. Claude Opus 4.6.



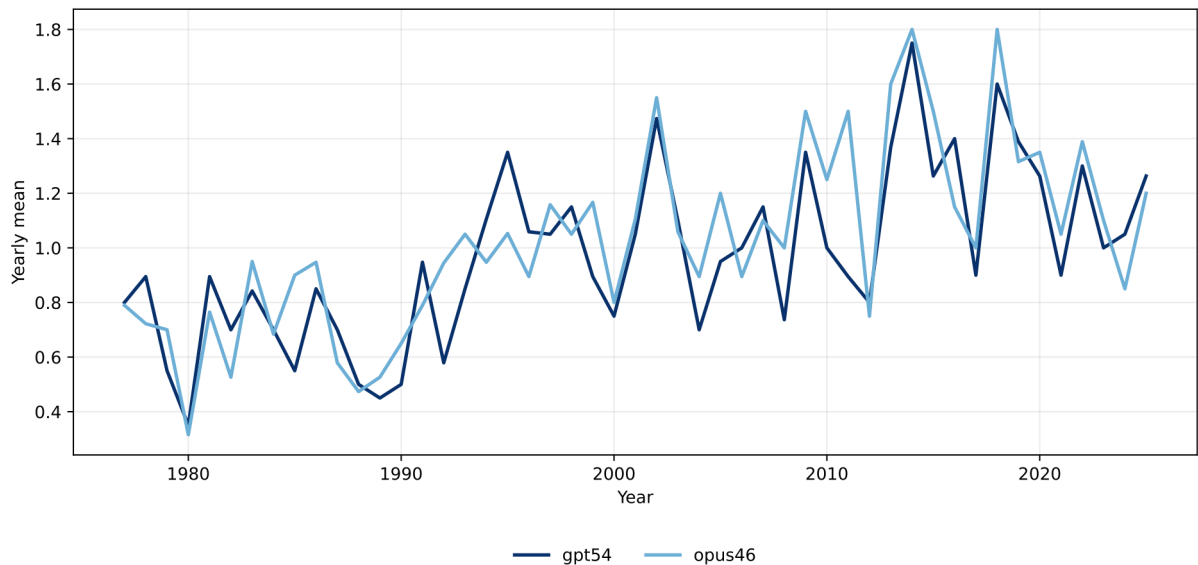
Notes: Each series is the yearly mean of the moral-scope index ($MSI = S + D$, range 0–6) on the full top-20 sample under one of the two coding models. GPT-5.4 is the primary coder, Claude Opus 4.6 is the cross-model replication on the identical sample. The mean year-by-year difference (Opus minus GPT) and the corresponding mean absolute deviation are reported in Table C.2. Sample: top 20 films per year ($n = 980$, 49 yearly means). Source: own coding.

Figure C.2: Mean Victim Level (S) Over Time: GPT-5.4 vs. Claude Opus 4.6.



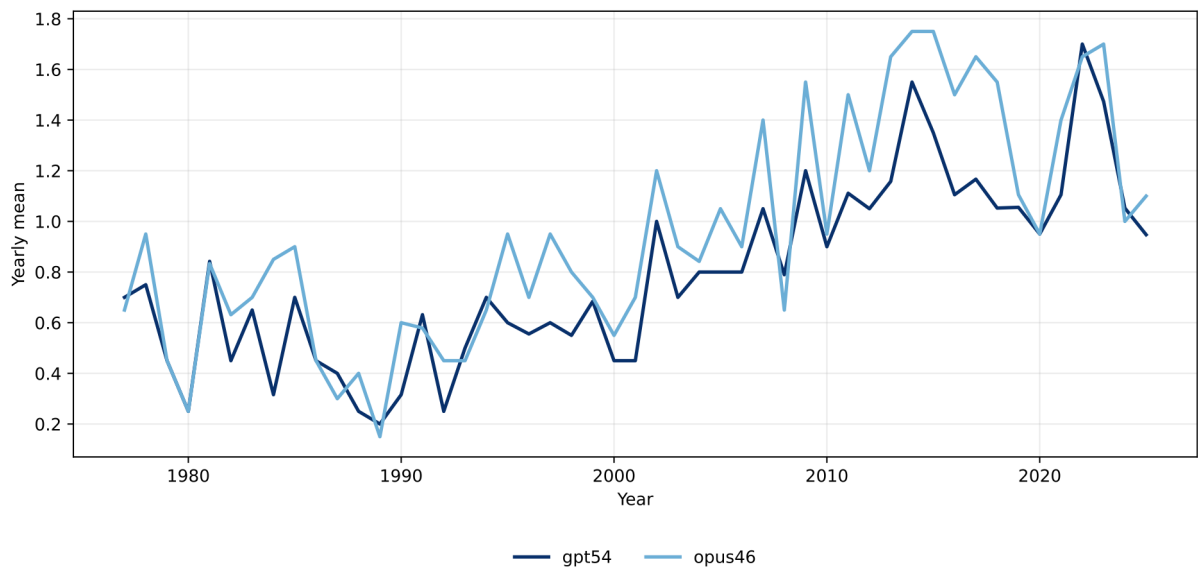
Notes: Each series is the yearly mean of the victim-level scale S (range 0–3) on the full top 20 sample under one of the two coding models. GPT-5.4 is the primary coder, Claude Opus 4.6 is the cross-model replication on the identical sample. The mean year-by-year difference (Opus minus GPT) and the corresponding mean absolute deviation are reported in Table C.2. Sample: top 20 films per year ($n = 980$, 49 yearly means). Source: own coding.

Figure C.3: Mean Hero-Victim Distance (D) Over Time: GPT-5.4 vs. Claude Opus 4.6.



Notes: Each series is the yearly mean of the hero-victim distance scale D (range 0–3) on the full top-20 sample under one of the two coding models. GPT-5.4 is the primary coder, Claude Opus 4.6 is the cross-model replication on the identical sample. The mean year-by-year difference (Opus minus GPT) and the corresponding mean absolute deviation are reported in Table C.2. Sample: top 20 films per year ($n = 980$, 49 yearly means). Source: own coding.

Figure C.4: Mean Conflict Scale (C) Over Time: GPT-5.4 vs. Claude Opus 4.6.



Notes: Each series is the yearly mean of the conflict-scale variable C (range 0–3) on the full top-20 sample under one of the two coding models. GPT-5.4 is the primary coder, Claude Opus 4.6 is the cross-model replication on the identical sample. The mean year-by-year difference (Opus minus GPT) and the corresponding mean absolute deviation are reported in Table C.2. Sample: top 20 films per year ($n = 980$, 49 yearly means). Source: own coding.

Year-by-year Difference Summary

Table C.2 summarizes the distribution of the year-by-year difference between the two runs (Opus minus GPT) for each of the three moral-scope components and for the composite MSI. All three scale variables are coded on an ordinal scale from 0 to 3. MSI is the sum of S and D and therefore ranges from 0 to 6. C is reported separately because it is not a component of MSI.

Table C.2: Year-by-Year Mean Differences (Opus – GPT), $n = 49$ years.

Variable	mean	mae	sd	min	max
S - victim level	0.151	0.172	0.163	-0.176	0.600
D – hero-victim distance	0.054	0.151	0.180	-0.297	0.605
C - conflict scale	0.162	0.186	0.178	-0.139	0.534
MSI - moral-scope index	0.204	0.284	0.295	-0.341	0.989

Notes: Values are differences of yearly means (Opus minus GPT). Positive values indicate that Claude Opus 4.6 codes higher on average than GPT-5.4 in that year. MAE is the mean absolute deviation of the yearly differences and captures the typical magnitude of disagreement between the two runs regardless of direction. The MSI is the sum of two ordinal components (victim level and hero-victim distance), so it accumulates their disagreements.

The mean differences are small relative to the scale of the variables (roughly 0.05 to 0.16 on scales that run from 0 to 3), and the MAE values are of the same order, which indicates that the systematic offset between the two models accounts for most of the

disagreement and that year-to-year noise around that offset is modest. The composite MSI accumulates the two component differences (victim level and hero-victim distance), which is why its mean and MAE are larger in absolute terms but not in relative terms.

The cross-model evidence in Figures C.1 to C.4 and Table C.2 converge on a single substantive conclusion. The absolute level at which any given variable is coded depends on the coding model. Claude Opus 4.6 produces slightly higher codings than GPT-5.4 on average. The temporal pattern of change, by contrast, does not depend on the coding model. The direction, timing, and relative magnitude of change across the sample period are similar under both coders. The trends documented in Chapter 6 are robust to the choice of LLM coder within the two-model design.

D. Additional Figures and Tables

This appendix collects supplementary figures and tables that support the analysis presented in the main text but are not essential to the core argument. The material included here serves mainly to document additional descriptive patterns, sample diagnostics, and exploratory results in a more detailed way. These outputs are reported separately to keep the main chapters focused while still making the underlying empirical material more transparent.

D.1 Additional Data Quality and Sample Diagnostics

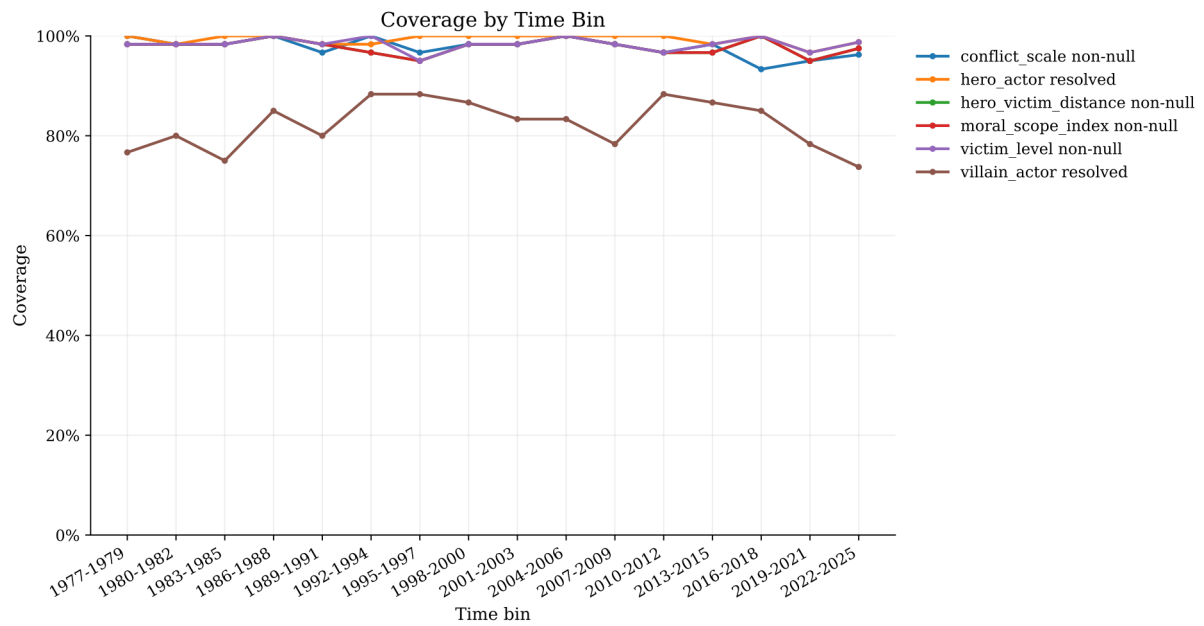
This appendix documents the data-quality diagnostics that underpin the analytic results in Chapter 6. The sample contains exactly 20 films per year for each of the 49 years from 1977 to 2025 ($n = 980$), and 99.9% of cases parsed cleanly through the structured-output schema described in Section 4.3 (81.0% clean, 18.9% parsed with non-blocking warnings, 0.1% skipped for absent synopsis). The three figures that follow document, in turn, the temporal stability of field-level coverage (Figure D.1), the overall missingness profile of the six core narrative-coding outputs (Figure D.2), and the distribution of the model's per-case confidence flag (Figure D.3). Table D.1 groups and explains the variables in the analysis dataset.

Table D.1: Variable Groups in the Analysis Dataset.

	Purpose	Main Variables
Sample	Define the observation unit and enable reproducible identification & matching across sources.	year; rank; title; stage4_row_key; tmdb_id; imdb_id_tmdb; tmdb_release_date; tmdb_match_confidence
Economics	Capture commercial success and market composition of each film.	worldwide_gross; domestic_gross; foreign_gross; domestic_pct; foreign_pct; domestic_rank; foreign_rank; domestic_foreign_ratio; rank_gap_*; *_missing flags
Inputs	Provide the textual basis for (i) narrative coding and (ii) keyword analysis, plus supporting evidence pointers.	<i>Narrative analysis (inputs):</i> synopsis_en; source_url; subtitle_available; subtitle_language; subtitle_path; subtitle_evidence <i>Keyword analysis (inputs)</i> tmdb_keywords
Outputs	Store (i) narrative coding results to test the hypotheses and (ii) keyword-analysis outputs, plus minimal quality signals.	<i>Narrative analysis (outputs):</i> hero_name; villain_name; victim_name; hero_actor; villain_actor; victim_level; hero_victim_distance; conflict_scale; moral_scope_index; confidence <i>Keyword analysis (outputs):</i> has_political_keyword; political_keyword_count; polshare_core; polshare_broad; political_any_core; political_any_broad

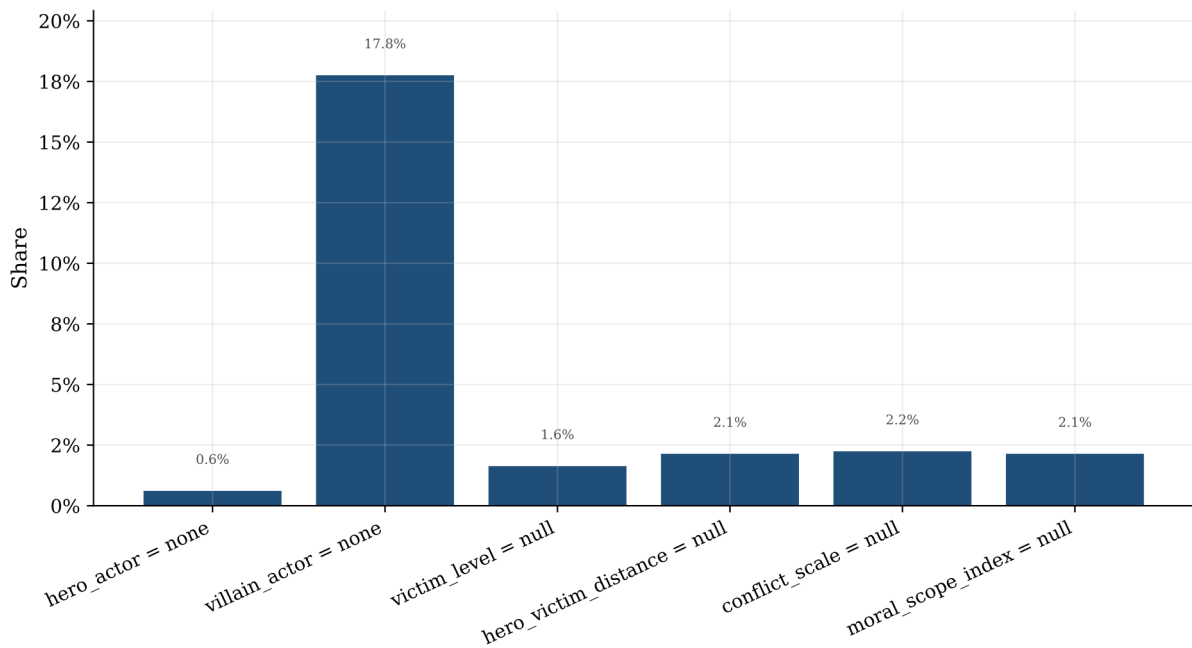
Notes: Variables are organized into four functional groups. *Sample* variables identify the observation unit and enable matching across external sources. *Economic* variables capture commercial performance, with separate measures for domestic, foreign, and worldwide gross plus derived shares and rank-based diagnostics. *Inputs* cover both narrative-coding inputs (English plot synopsis from OMDb with TMDb and Wikipedia fallbacks, subtitle evidence from OpenSubtitles) and keyword-analysis inputs (TMDb keyword metadata). *Outputs* include both the keyword-based political-content indicators (Stage 3) and the LLM-coded narrative variables: hero, villain, and victim names, hero and villain actor types, the three ordinal scale variables (S, D, C), and the Moral Scope Index (see Section 2.3). Sample: top 20 highest-grossing films worldwide per calendar year, 1977–2025 (n = 980). Source: own dataset, constructed from Box Office Mojo, TMDb, OMDb, and OpenSubtitles.

Figure D.1: Coverage of Narrative-Coding Output, 1977–2025.



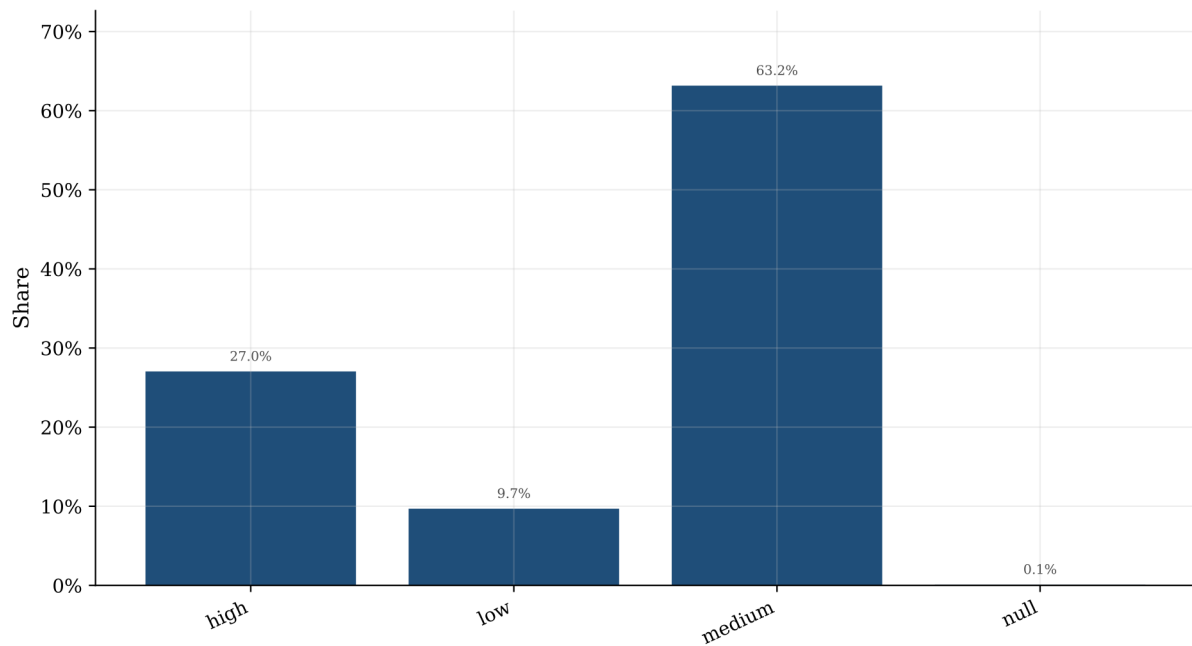
Notes: The figure plots, for each of the six core narrative-coding outputs, the share of films per 3-year bin with a usable value. For the two actor-typing fields (hero_actor, villain_actor) "usable" means the model resolved the role to one of the allowed actor categories defined in Chapter 4, for the four ordinal scales (S, D, C, MSI) it means the model returned a non-null integer in the 0–3 (or 0–6 for MSI) range. Bins of three years smooth year-to-year noise without obscuring longer-run shifts. The relevant pattern for the analysis in Chapter 6 is that all six series remain high (mostly above 90%) and roughly flat across the 49-year window. Field-level coverage therefore does not change in a way that would mechanically generate the trends documented in Chapter 6. The villain_actor series sits visibly lower than the others throughout the period, which is consistent with the overall missingness pattern in Figure D.2 and with the discussion of villain identification in Chapter 4. Sample: 980 films, 20 per year, 1977–2025. Source: own coding.

Figure D.2: Field-Level Missingness and Unresolved Rates Across the Full Sample.



Notes: The figure plots, for each of the six core narrative-coding outputs, the share of films across the full sample for which the field is missing in the analysis dataset. For the two actor-typing fields (hero_actor, villain_actor) "missing" means the model returned none, i.e. could not assign one of the allowed actor categories. For the four ordinal scales (S, D, C, MSI) "missing" means the model returned null, following the unresolved-labelling policy described in Section 4.3. With the exception of villain_actor (17.8%), all fields show missingness rates below 3%, indicating that the analytic measures used in Chapter 6 rest on near-complete data. The elevated villain_actor rate is consistent with the cross-model agreement pattern reported in Section 4.4.2 and reflects the inherent ambiguity of identifying a single antagonist in films where the source of conflict is diffuse, structural, or absent. Sample: 980 films, 1977–2025. Source: own coding.

Figure D.3: Distribution of Declared Model Confidence Across the Full Sample.



Notes: The figure plots the share of films at each level of the per-case confidence flag returned by GPT-5.4 as part of the structured output schema defined in Section 4.3. Four levels are possible: high (27.0%), medium (63.2%), low (9.7%), and null (0.1%, where the model failed to return a confidence label). This distribution closely matches the confidence distribution of the 10% manual-review subsample reported in Appendix C.2 (33% high, 58% medium, 9% low), confirming that the review subsample is representative of the full sample on this dimension. Read together with Table C.1, the figure also motivates the analytic decision to retain low-confidence cases as unresolved rather than to drop or impute them: low-confidence cases are coded plausibly far less often than high- or medium-confidence cases (44.4% implausibility rate vs 0% and 17.5%), so they are precisely the cases where the coding signal is weakest. Sample: 980 films, 1977–2025. Source: own coding.

D.2 Additional Narrative-Coding Descriptives

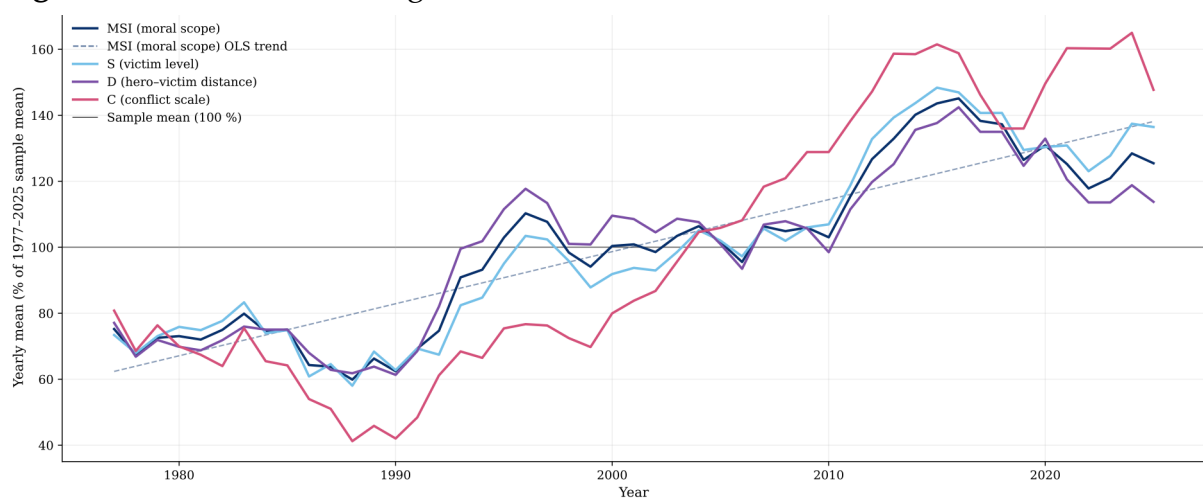
Appendix D.2 reports three views of the narrative-coding output that complement Chapters 6 and 7. Figure D.4 plots all four indices on a common normalized scale, providing a single-frame summary of the trends estimated in Table 3. Figures D.5 and D.6 trace compositional shifts in the hero and villain actor-type taxonomies, two categorical fields whose variation is suppressed once the ordinal scales are constructed. Figures D.7 and D.8 show how the ordinal scale variables S and D distribute over time, the geographic-scope variable C is shown in Figure D.9. Table D.2 reports year-residualised correlations among the four narrative variables, isolating within-year covariation from the common time trend.

Table D.2: Pairwise Correlations Among Narrative-Scope Measures

Pair	(1) Film level	(2) Yearly means	(3) Film level, residualized	(4) Yearly means, residualized
S – D	0.682	0.816	0.671	0.670
MSI – C	0.650	0.796	0.628	0.541
S – C	0.607	0.806	0.580	0.546
D – C	0.587	0.702	0.570	0.433

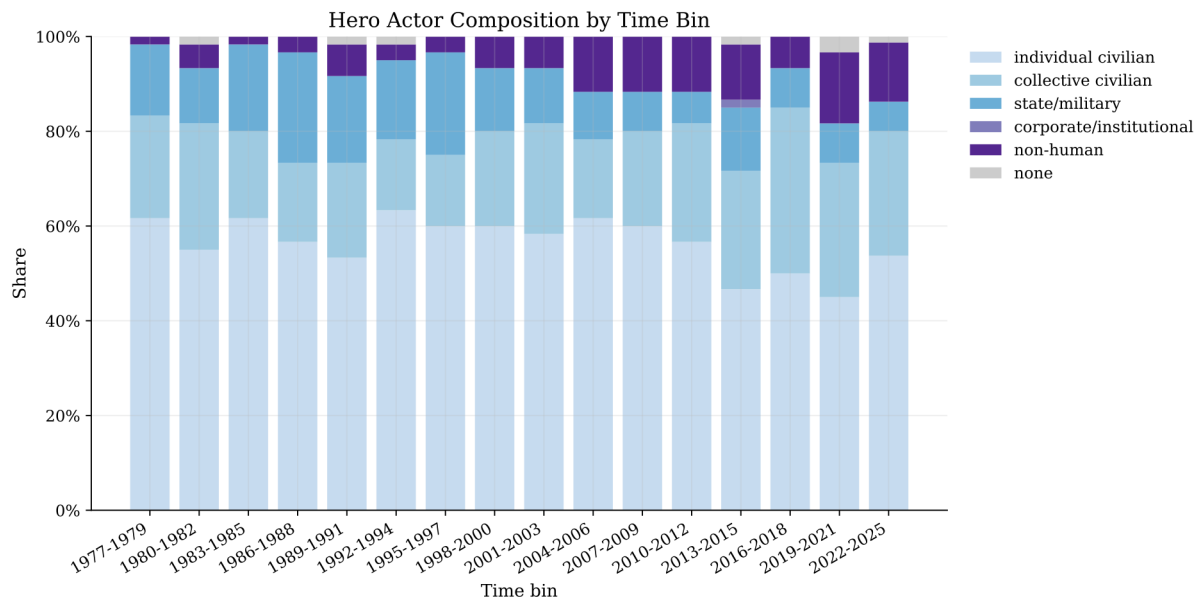
Notes: Entries are Pearson correlations. Film-level correlations are computed across individual films. Yearly-mean correlations are computed across annual means. Residualized correlations are computed after regressing each variable on a linear year trend and correlating the residuals. MSI = Moral Scope Index, S = Victim Level, D = Hero-Victim Distance, C = Geographic Conflict Scale. MSI is constructed from S and D high correlations are thus mechanical. Source: own estimation.

Figure D.4: Normalized Long-Run Trends of the Four Narrative Indices, 1977–2025.



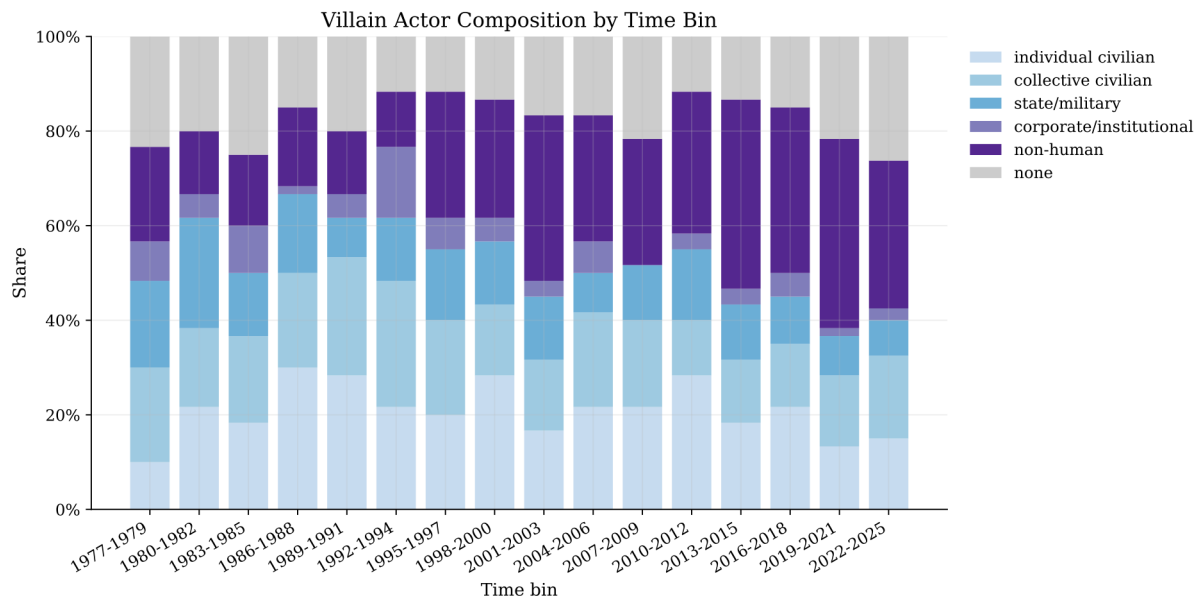
Notes: Each series is the yearly mean of the indicated index, rescaled so that the 1977–2025 sample mean equals 100. The dashed line is the OLS trend in MSI on year on the rescaled series, trend coefficients on the original 0–6 (MSI) and 0–3 (S, D, C) scales are reported in Table 3. Note that MSI = S + D by construction, so the MSI line is mechanically a function of the S and D lines and should not be read as independent evidence. Sample: top 20 films per year (n = 980, 49 yearly means). Source: own coding using GPT-5.4.

Figure D.5: Hero Actor-Type Composition Across 3-Year Time Bins, 1977–2025.



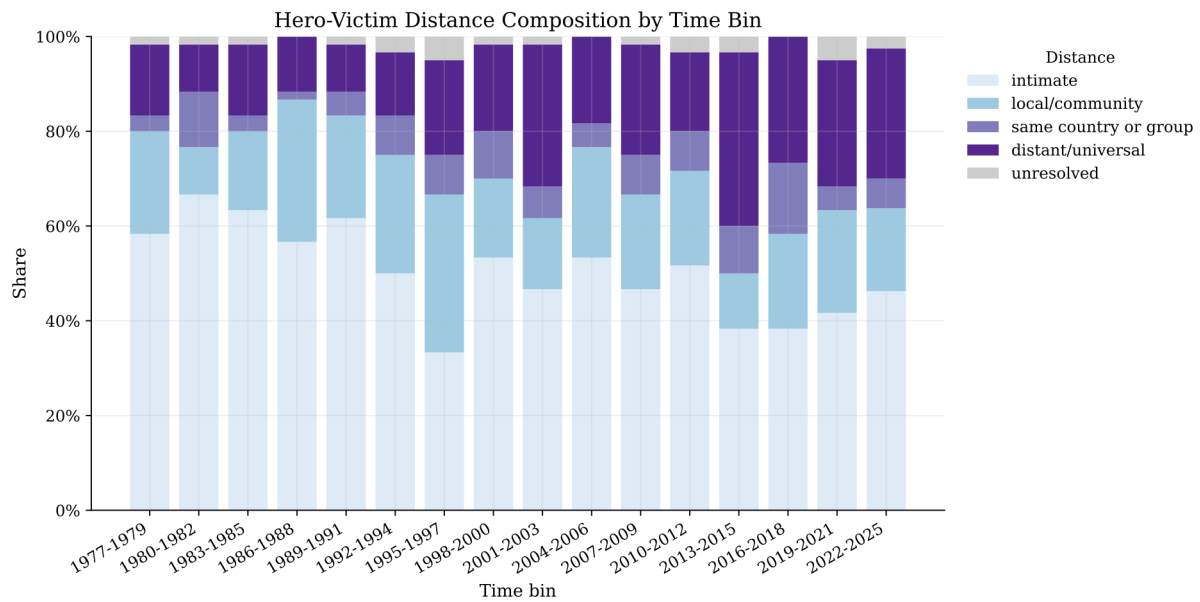
Notes: Each bar shows the share of films in a 3-year bin assigned to each hero-actor category. Categories are mutually exclusive and exhaustive within the matched sample: individual civilian, collective civilian, state or military, corporate or institutional, non-human, and none (no clear hero). Bins are aligned to the calendar years shown on the horizontal axis, the final bin covers 2022–2025 and is partial. Sample: top 20 films by worldwide box office per year ($n = 980$). Source: own coding using GPT-5.4 on TMDb plot metadata.

Figure D.6: Villain Actor-Type Composition Across 3-Year Time Bins, 1977–2025.



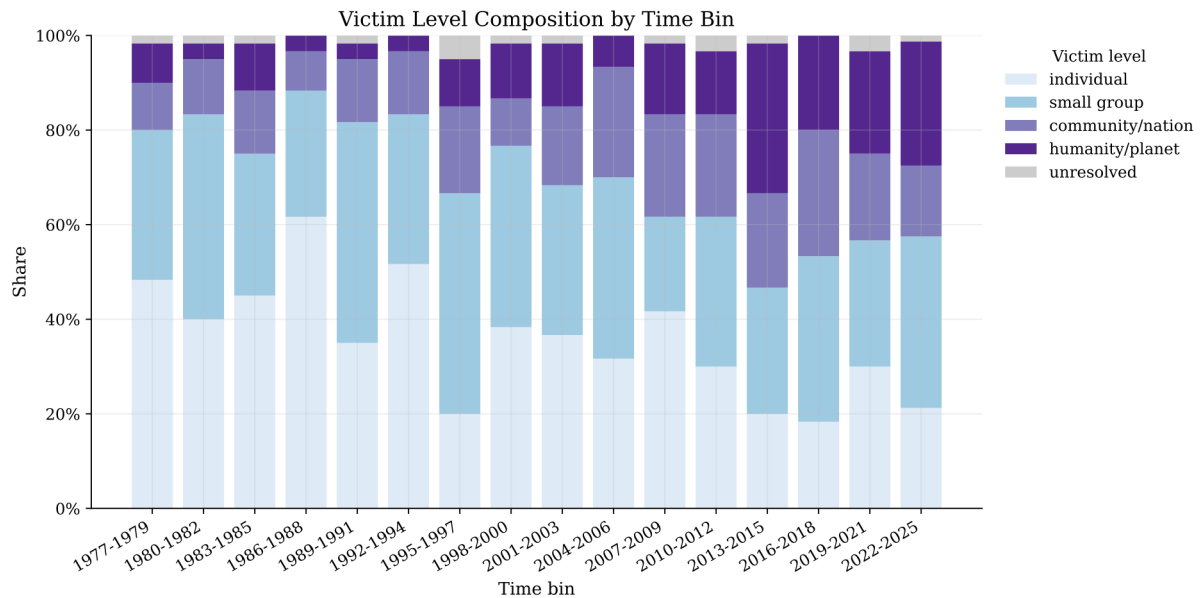
Notes: Each bar shows the share of films in a 3-year bin assigned to each villain-actor category. Categories are the same as in Figure 5, with "none" denoting films coded as having no clear villain (17.8% of the matched sample, see Figure D.2). Sample: top 20 films per year ($n = 980$). Source: own coding using GPT-5.4 on TMDb plot metadata.

Figure D.7: Hero-Victim Distance Composition Across 3-Year Time Bins, 1977–2025.



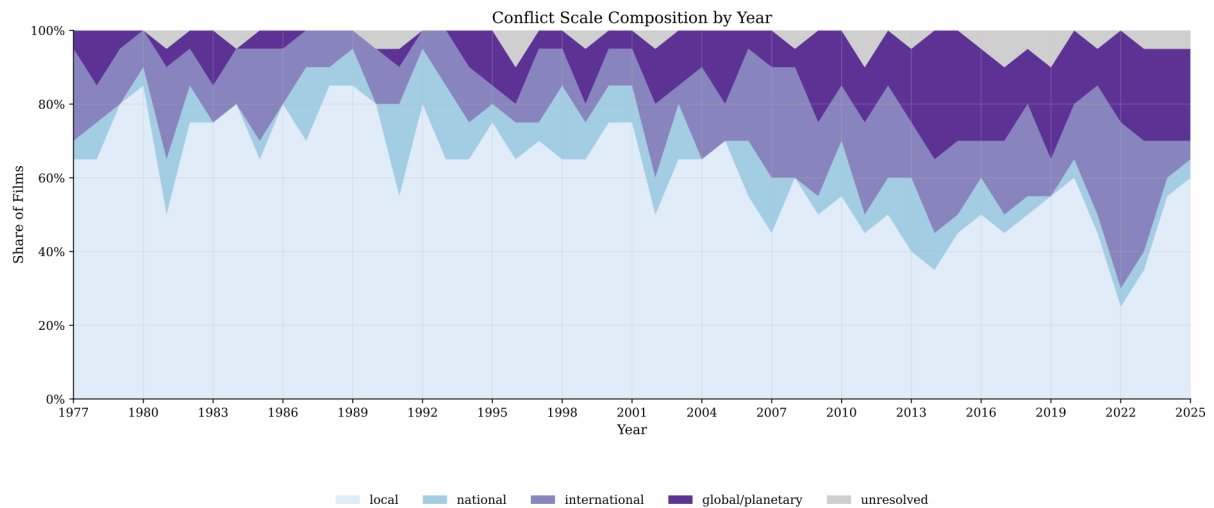
Notes: Each bar shows the share of films in a 3-year bin assigned to each level of hero-victim distance D . Levels are intimate ($D = 0$), local or community ($D = 1$), same country or group ($D = 2$), and distant or universal ($D = 3$). The unresolved share corresponds to films for which the available evidence was insufficient to assign a code (see Section 4.3). The ordinal scale used in Chapters 5–6 is constructed by dropping the unresolved cases and treating D as 0–3. Sample: top 20 films per year ($n = 980$). Source: own coding using GPT-5.4.

Figure D.8: Victim-Level Composition Across 3-Year Time Bins, 1977–2025.



Notes: Each bar shows the share of films in a 3-year bin assigned to each level of victim level S . Levels are individual ($S = 0$), small group ($S = 1$), community or nation ($S = 2$), and humanity or planet ($S = 3$). The unresolved share corresponds to films for which the available evidence was insufficient to assign a code (see Section 4.3). The ordinal scale used in Chapters 5–6 is constructed by dropping the unresolved cases and treating S as 0–3. Sample: top 20 films per year ($n = 980$). Source: own coding using GPT-5.4.

Figure D.9: Conflict Scale Composition by Year, 1977–2025.



Notes: Yearly composition of the coded conflict scale C ($0 = \text{local}$, $1 = \text{national}$, $2 = \text{international}$, $3 = \text{global/planetary}$) for the 20 highest-grossing films worldwide per calendar year, 1977–2025. Each vertical slice shows the share of films in year t falling into each C category, ordered from local at the bottom to global/planetary at the top. The grey "unresolved" band contains films in which C could not be coded ($n = 22$). A linear fit of the yearly mean C across the 49 years yields $\beta = +0.019$ per year ($p = 5.8 \times 10^{-11}$, $R^2 = 0.60$), the strongest linear trend among the three coded relations.

E. Declaration of Independent Work

I hereby declare that I have written this thesis without any help from others and without the use of documents and aids other than those stated above. I have mentioned all used sources and cited them correctly according to established academic citation rules. I am aware that otherwise the Senat is entitled to revoke the degree awarded on the basis of this thesis, according to article 36 paragraph 1 letter o of the University Act from 5 September 1996.

20.05.2026, Aarau

Cédric Rötheli
